The LQD Model: Unconstrained Coordinates for Arbitrage-Free Smiles

A density-first lecture on volatility fitting in the exponential-tail class

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Abstract

A volatility smile is a probability distribution seen through the Black formula. Most fitters nevertheless parametrize the smile directly and then police, with penalties, the awkward nonlinear conditions under which their curve still corresponds to a genuine distribution. This lecture develops the Vol-Fitter's *LQD* (log-quantile-density) model the other way around, as the answer to a coordinate problem: among all the charts one can draw on arbitrage-free slices — price, implied volatility, distribution function, density, quantile — find one in which a class of laws rich enough for equity smiles becomes *all of coordinate space*, so that the optimizer can roam freely and every point it visits already prices from a genuine law. The answer is to model the logarithm of the quantile density of the log-forward return as two universal boundary terms plus a Legendre expansion, $\log q(u) = -\log u - \log(1-u) + g(u)$. The class this chart covers is made precise (mean-one laws with exponential tails; a density proposition says in what sense it is everything, and what — atoms, gaps, Gaussian tails — lies outside it). Positivity of the density is structural, a single scalar shift enforces the martingale condition, and the two endpoint values of g identify both the last finite moments of the distribution and its Lee wing slopes; the one inequality that remains, $A_R < 1$, the price of a finite forward, is absorbed by a logistic coordinate, so the production solve runs over genuinely unconstrained $\mathbb{R}^d$. From this construction we derive pricing as one cumulative integral, exact closed-form ATM level/skew/curvature handles with an orthogonal chart around them, the calibration objective, an analytic residual Jacobian obtained from a clean cancellation identity, and soft convex-order control across expiries — read off the same upper-share ledger that prices the calls. What the continuous construction proves, a numerical certificate must inherit: a randomized strike-space audit bounds the implementation's residual arbitrage below 10^{-8} in normalized price, and a stability study measures — rather than asserts — how far the tail parameters are identified by quotes. Every figure is generated by the production calibrator on deterministic synthetic benchmarks: a ten-parameter fit to an SPX-like strip reaches a maximum error of 1.341 volatility basis points, and a seventeen-parameter fit recovers an asymmetric bimodal event density to 2.444 vol basis points. Market-fit evidence lives in the backtest reports; what this lecture establishes is the machine itself.

Contents

1 The shape of the problem

Here is a failure mode every production smile fitter meets. You parametrize implied volatility with some pleasant curve, fit seven liquid strikes, and the residuals look excellent. Then a risk system differentiates your curve twice — because the second strike derivative of a call *is* the risk-neutral density, as we recall below — and finds it negative between two quotes. Your smooth, well-fitting smile is quoting a butterfly you would pay to be lifted on: free money for the counterparty, a phantom probability of -2% for the risk report. Nothing in the fit was wrong numerically. The parametrization simply did not know that a smile is not free to be any curve.

The standard remedy is to keep the pleasant curve and add penalties: evaluate the density (or its proxy) on a grid, punish negativity, and hope the optimizer lands inside the feasible set. This works, mostly, but it has a characteristic cost profile. The constraint is *global* and *nonlinear* in the parameters, so it must be re-checked at every strike and every iterate; the optimizer spends its time crawling along the boundary of the feasible set, exactly where the penalty gradients fight the data gradients; and when the market is sparse or stressed — precisely when you most need the fitter — the boundary is where the solution wants to live.

The LQD model spends none of that effort, because it asks a different question first.

> *The set of arbitrage-free slices is a well-defined mathematical object. In which coordinates is that set trivial — all of coordinate space, no boundary, nothing to police?*

This lecture is the answer worked out in full, and the sense in which it is a lecture is worth one sentence: we will build the model from first principles, prove the small number of facts that carry it, and keep the numerical plumbing in clearly marked asides, so that a reader who knows the Black formula and one year of probability can reconstruct everything. The destination is fixed by production, and it is protected by the following invariants.

Invariant

1. Every admissible parameter vector prices from a genuine non-negative density: butterfly-freedom is structural, never a penalty (proposition 2).

2. The only hard admissibility condition is the single inequality $A_R < 1$ — a finite forward. In the production chart it is not even that: the logistic coordinate $A_R = \operatorname{expit}(\rho)$ makes the wall unreachable, and what remains at the numerical boundary is an explicit overflow budget, not a constraint (sections 4 and 9).

3. The martingale normalization is one scalar shift; it changes neither the density's shape, nor its tails, nor its positivity (section 7).

4. The ATM handles $(\sigma_0, s_0, \kappa_0)$ are exact closed-form functionals of the slice — the same three coordinates the graph extrapolator of Note 14 propagates (section 8).

A running example. Lectures profit from a recurring character. Ours is a deterministic, SPX-like strip of 24 quotes at the expiry $\tau = 0.75$, sampled from a skewed SSVI-shaped curve (appendix C) with a reproducible sub-two-basis-point microstructure ripple, and fitted *once* by the production calibrator with a nine-mode LQD slice. Almost every figure in this note draws that one fitted slice from a different angle; its full case file, with numbers, is section 12.1. A second character — a bimodal event distribution — appears when we need to stress the basis (section 12.2).

What this note is not. The examples are *synthetic approximation benchmarks*: they measure how well the parametrization can reproduce a known clean curve, not how well it fits noisy market quotes. Market-fit evidence — RMS against bid–ask NBBO quotes, in and out of sample, against an SVI-JW baseline — lives in the backtest harness (`backend/backtest/`) and its reports.

2 What a smile really is

2.1 Normalized units, and the three clocks

Fix one expiry. All the securities we price are functions of the terminal spot S, and everything deterministic — discounting, forwards, carry — can be divided out before modelling begins. We therefore work with the *log-forward return*

$$X = \log \frac{S}{F}, \qquad Y = e^X = \frac{S}{F}, \tag{1}$$

with F the forward to the expiry, under the forward measure of that expiry. Strikes are quoted in log-moneyness $k = \log(K/F)$, and the *normalized call* is

$$C(k) = \mathbb{E}\Big[(e^X - e^k)^+\Big], \qquad \mathbb{E}[e^X] = 1, \tag{2}$$

where the second equality — the martingale, or forward, condition — says that the forward really is the mean of the terminal spot. A dollar quote $C^{\$}$ converts to these units by $C = C^{\$}/(DF)$ with D the discount factor; nothing else about rates or carry will appear again. Deterministic carry is a standing assumption of the whole note; it is what legitimizes working slice by slice under each expiry's forward measure, and it is revisited when slices are coupled in section 11.

One symbol of time is enough, but it must be the right one. Production distinguishes the *expiry date* from its *calendar year fraction* and from the *active variance time* — calendar time dilated by scheduled events (Note 11). Everything in this note that multiplies a variance, scales a vega, or annualizes a handle uses the active time, written

$$\tau \ = \ \text{active variance time to expiry} \qquad \text{(the calibrator's \texttt{prepared.tau})}, \tag{3}$$

which equals the calendar fraction whenever the event clock is off. Total implied variance at log-moneyness k is $w(k) = \sigma^2(k)\,\tau$, with σ the Black implied volatility. A *volatility basis point* (vol bp) is 10^{-4} in volatility. When several expiries appear at once we index them $\tau_1 < \tau_2 < \cdots$; a single unlabelled slice is always "the" expiry.

2.2 Breeden–Litzenberger, and the slice as a law

The reason a smile cannot be an arbitrary curve is one integration by parts away. Write the call as a function of the normalized strike $y = e^k$; when Y has a density f_Y,

$$\frac{\partial C}{\partial y} = -\,\mathbb{P}(Y > y), \qquad \frac{\partial^2 C}{\partial y^2} = f_Y(y) \ \geq \ 0. \tag{4}$$

This is the Breeden–Litzenberger identity [1]: the call curve determines the distribution, and conversely every property of the call curve that a static portfolio can monetize is a property of the distribution. A *butterfly* — long one call at $y-h$, long one at $y+h$, short two at y — costs $C(y-h) - 2C(y) + C(y+h)$, and by (4) its price per h^2 converges to the density at y. A negative butterfly price is exactly a negative density.

Definition 1 (Arbitrage-free slice)**.** A *slice* is a call curve $C(k)$ of the form (2) for some law of X with $\mathbb{E}[e^X] = 1$. Equivalently (for laws with a density): C is decreasing and convex in the strike $y = e^k$, with $C \to 1$ as $y \to 0$ and $C \to 0$ as $y \to \infty$, slope ≥ -1, and the boundary derivative condition $C'(y) \to -1$ as $y \to 0^+$ — the slope at zero strike *is* $-\mathbb{P}(Y > 0)$, so anything shallower than -1 there would be default mass at zero, which a law on $(0, \infty)$ excludes.

The space of slices, then, is really the space of mean-one probability laws on $(0, \infty)$. That is the object we must parametrize, and fig. 1 shows our running example wearing both descriptions at once: a convex call curve on the left, and on the right the density recovered by pricing actual finite-width butterflies on the fitted slice.

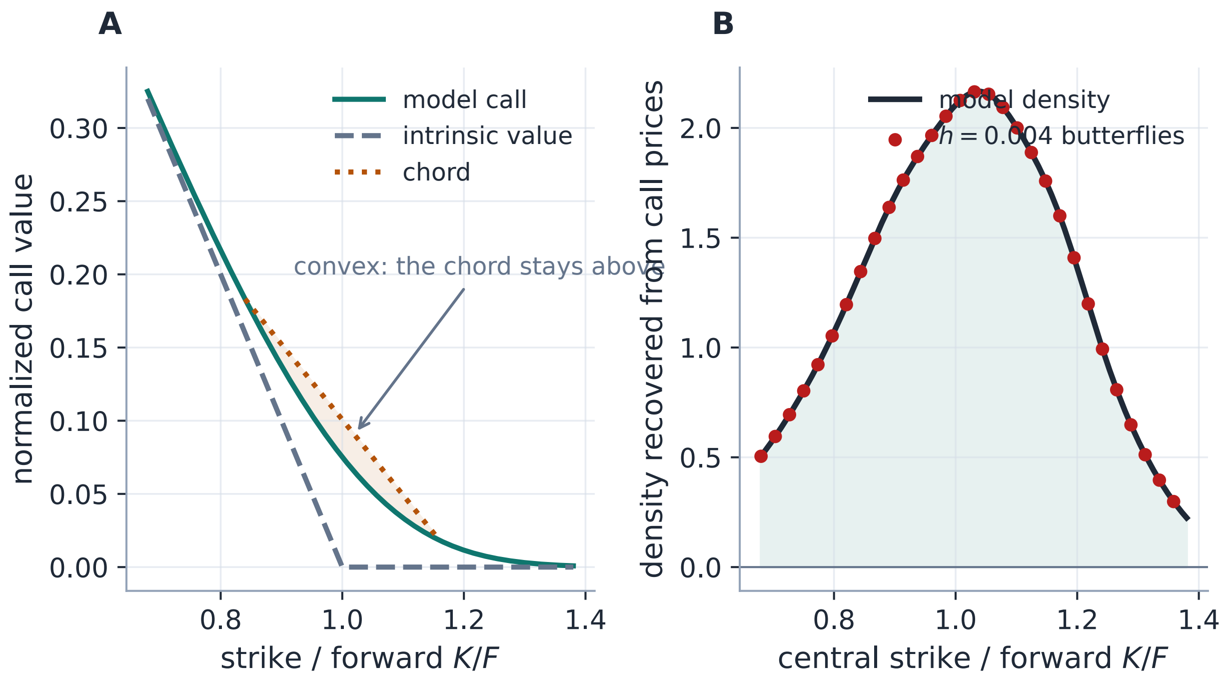

Figure 1: The running SPX-like fit as a tradable object. Panel A: the normalized call curve is decreasing and convex — every chord sits above the curve, so every butterfly has non-negative price. Panel B: discrete butterflies of half-width $h = 0.004$, priced off the fitted slice and divided by h^2, recover the model density to a relative 0.0182 (the minimum butterfly value on the grid is $0.224428 > 0$). Butterfly-freedom is not something this fit achieved; it is something the parametrization cannot lose.

2.3 The notation ledger

Because this lecture insists on few symbols, here they all are. Each is defined again where it first works for a living; none is ever reused with a second meaning.

$X,\ Y = e^X$	log-forward return, gross return	$u,\ z = \log \frac{u}{1-u}$	rank (percentile), log-odds
$k,\ C(k),\ P(k)$	log-moneyness; call, put	$Q,\ q = Q'$	quantile of X; quantile density
$w,\ \sigma,\ \tau$	total variance $\sigma^2\tau$; vol; active time	$g,\ \theta$	smooth part of $\log q$; coefficients
$B(k, w)$	normalized Black call	$A_L,\ A_R$	endpoint (tail) scales $e^{g(0)}, e^{g(1)}$
μ	martingale shift	G	upper share $\int e^Q$ above a rank
$\sigma_0,\ s_0,\ \kappa_0$	ATM vol, skew, curvature	$\beta,\ \psi,\ p$	Lee slope, Lee map, moment order
$\lambda, r;\ c, \gamma$	ridge; barrier centre/steepness	H, J, U, V, α, ξ	the ATM-orthogonal chart

Table 1: Every symbol in the note. Φ and φ are the standard normal CDF and PDF. One differentiation convention holds throughout: a *prime* is the derivative of a function of one variable (C', w', σ'); a *subscripted* ∂ is a partial derivative of a function of several ($\partial_k B$, $\partial G/\partial\theta$). Symbols local to one appendix (the SSVI oracle's ϑ, ρ, ϕ) stay in that appendix.

3 Six charts for one space

We are hunting for coordinates on the space of mean-one laws. A good way to appreciate the answer is to price the alternatives first — this is a shopping trip, and the currency is *constraints left over*.

The price chart. Parametrize $C(k)$ itself. The constraint set definition 1 is a convex set (with the normalization $C(0^-) = 1$ it is a slice through a cone, not the cone itself), and its boundary is a continuum of *linear* functional inequalities: monotonicity and convexity at every strike, plus the endpoint conditions. Linear or not, enforcing a continuum of inequalities for a flexible family means a grid of constraint rows and an optimizer that lives on the boundary. Nothing is free.

The volatility chart. Parametrize $\sigma(k)$ — what every trader quotes, and what SVI does. The density is now a *nonlinear* second-order differential expression in $w(k) = \sigma^2(k)\tau$ (Gatheral's g-function [3, 6]); non-negativity is an implicit, parameter-global boundary that no useful family satisfies identically. This chart is the most convenient to *quote* and the most expensive to *police* — the note's opening failure mode lives here.

The distribution chart. Parametrize the CDF F_X. One constraint survives: monotonicity, again at every point. Closer.

The density chart. Parametrize $f_X \geq 0$ directly. Positivity of a flexible parametric curve is still a pointwise constraint (unless one squares or exponentiates, of which more below), and two integral constraints remain: total mass one, and the martingale $\int e^x f_X(x)\,\mathrm{d}x = 1$. To be fair to this chart: exponentiating ($f_X \propto e^\phi$) buys positivity for free, a normalizing constant absorbs the mass, and a translation of X can absorb the martingale condition — an exponential density family can be pushed most of the way LQD goes. What it cannot buy so cheaply is the rest of this lecture: quantile-side pricing as one cumulative integral, endpoint values that *are* the tail scales, and the upper-share ledger. The comparison that matters is numerical, and it is made where the machinery exists to make it (section 7.2 and appendix B), not by counting constraints.

The quantile chart. Parametrize the quantile function $Q = F_X^{-1}$ on the rank $u \in (0,1)$. One constraint: Q strictly increasing. And here the structure finally cooperates, because monotonicity — unlike convexity or positivity-of-a-difference — is destroyed by *one* derivative, not two.

The free chart. Let $q(u) = Q'(u)$ be the *quantile density*. Then Q increasing $\iff$ $q > 0$ $\iff$ $\log q$ is *any* real function. Modelling $\log q$ removes the last pointwise constraint; what remains are one integrability condition (so that the martingale integral converges) and one scalar normalization (to set its value to one) — and, remarkably, we will see in sections 4 and 7 that these cost a single inequality and a single additive shift. Table 2 tallies the trip, and fig. 2 shows the same fitted slice in four of the charts: the constraints that are visibly at work in the first three panels are simply *absent* in the fourth.

Chart	Object	Constraints left over	Type
Price	$C(k)$	decreasing, convex, endpoints	pointwise, global
Volatility	$\sigma(k)$	$f \geq 0$ as a nonlinear ODE expression	implicit, global
Distribution	F_X	increasing	pointwise
Density	f_X	$f \geq 0$; mass $= 1$; $\mathbb{E}[e^X] = 1$	pointwise + 2 integrals
Quantile	$Q(u)$	increasing	pointwise
Log quantile density	$\log q(u)$	$A_R < 1$; one scalar shift	*one inequality*

Table 2: The shopping trip. Each chart can describe the same laws; the constraint column is what a *flexible parametric family drawn in that chart* must police to stay arbitrage-free. Only the last reduces the boundary to a single scalar inequality (absorbed entirely by the logistic coordinate of section 5.1), with the normalization a shift rather than a constraint. The table is a statement about parametrization cost, not a claim that the finite family reaches every law — proposition 1 below says exactly which laws it reaches.

What the free chart covers. Freedom has a scope, and it should be stated rather than implied. A finite-order LQD slice has a strictly positive, continuous density on an interval, with exponential tails in X. It therefore cannot *exactly* represent atoms (a pinned settlement print), bounded support, interior intervals of zero density, default mass at zero, or Gaussian and faster-than-exponential tails. What it can do is approximate: the following proposition is the honest version of "coordinates for the space of smiles".

Proposition 1 (Density of the family)**.** *Let X be a mean-one law ($\mathbb{E}[e^X] = 1$) whose quantile function Q is differentiable with $\log Q'(u) + \log u + \log(1-u) =: g^*(u)$ extending continuously to $[0,1]$ — equivalently, a law with a positive continuous density and exponential tails of scales $A_L = e^{g^*(0)} > 0$ and $A_R = e^{g^*(1)} < 1$. Then the LQD family is dense at X: writing g_N for the degree-N polynomial approximations of g^* furnished by Weierstrass' theorem, every g_N is itself an admissible slice for N large enough, $Q_N \to Q$ uniformly on compact subsets of $(0,1)$ with uniform control of the tails, and consequently the normalized call curves converge uniformly in k and the laws converge in Wasserstein-p distance for every p below the critical moment $1/A_R$. Conversely, a law with an atom, a density gap, or super-exponential tails is not in the family for any finite N, and for atoms and gaps not even in its closure under these metrics with admissible limits: the family's closure is exactly the exponential-tail class.*

Proof sketch. Uniform approximation $g_N \to g^*$ on $[0, 1]$ (Weierstrass; the Legendre partial sums of a continuous g^* after Cesàro smoothing serve) keeps $g_N(1) \to g^*(1) < 0$, so $A_R^{(N)} < 1$ eventually: admissibility is an *open* condition, which is what makes the family stable under approximation. Uniform closeness of g gives uniform closeness of $\log Q'$ hence of Q up to the martingale shift, which converges because the normalization integral does (dominated convergence with the exponential envelope). Call prices are Lipschitz functionals of Q in the relevant metric via the upper-share representation of section 7; Wasserstein-p convergence for $p < 1/A_R$ follows from uniform quantile convergence plus the uniform exponential tail bound. An atom is a jump of Q; a gap is a flat of Q; both are excluded uniformly by any finite g, and a Gaussian tail forces $g(u) \to -\infty$ at the endpoint, off the chart. □

The proposition earns the title's word "unconstrained" and prices its word "class": within the exponential-tail class the coordinates cover everything, and what falls outside is named, not hidden. For desk use the exclusions matter mostly as approximation targets — section 12 fits a genuinely bimodal density and measures what a finite N delivers — and as honest limits (section 14 returns to atoms and spiky densities).

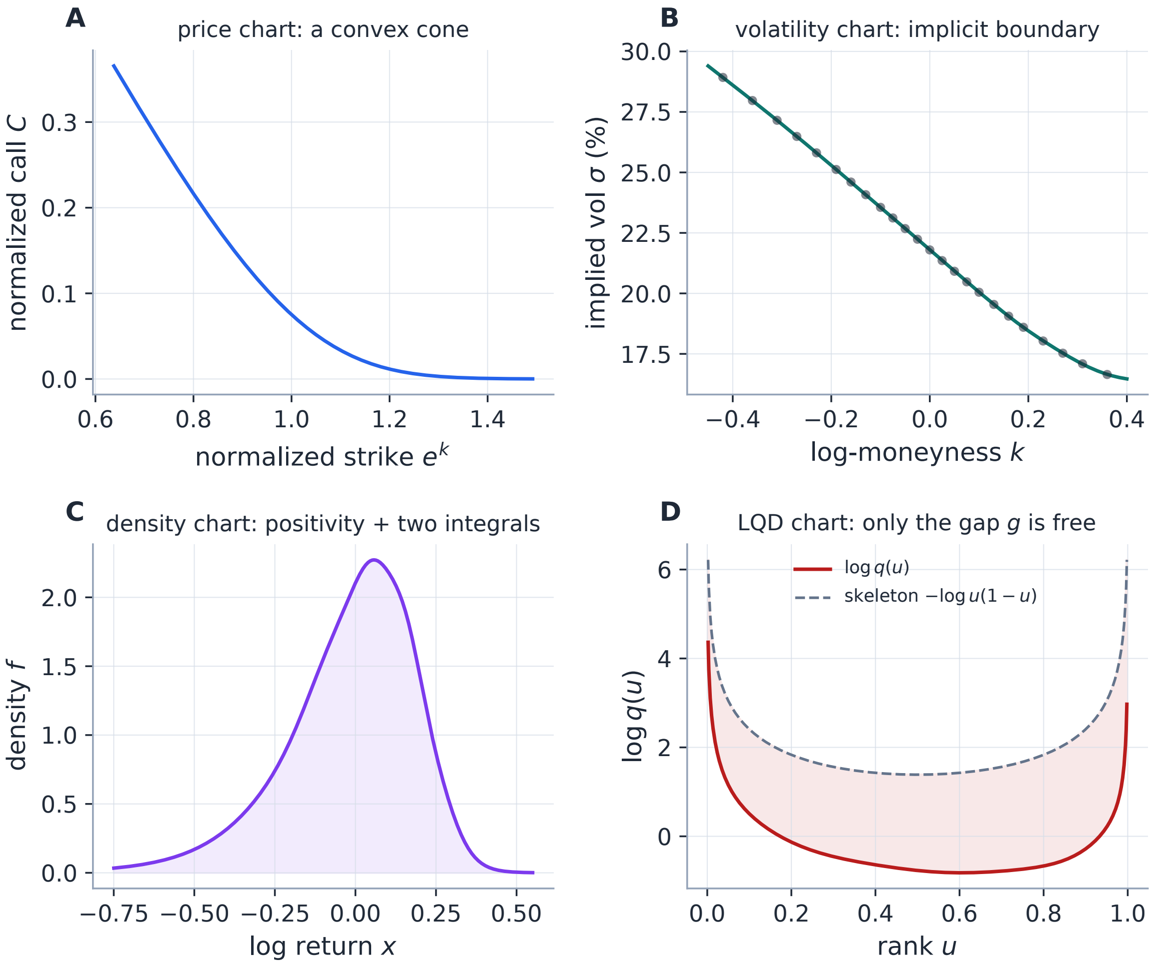

Figure 2: One fitted slice — the running SPX-like example — in four coordinate charts. A: the price chart, where no-arbitrage is a convex cone. B: the volatility chart, where it is an implicit nonlinear boundary (the quotes ride the fitted curve). C: the density chart, where it is positivity plus two integral constraints. D: the LQD chart of definition 2: the classical U of $\log q$, diving to $+\infty$ at both ends along the universal skeleton $-\log u - \log(1-u)$ (dashed). The skeleton is the same for every slice; the model's entire freedom is the shaded gap g between the two curves, and *any* smooth gap — shifted, bent, or oscillating — is another arbitrage-free slice, provided only that the single inequality $A_R < 1$ holds at the right edge. The fit's martingale check is $|\mathbb{E}[e^X] - 1| = 1.11 \times 10^{-16}$.

3.1 The model in one equation

It remains to choose a basis for $\log q$, and the choice is forced by two requirements: the tails must be *right by construction* (next section), and the body must be flexible, well-conditioned, and cheap. The LQD answer splits $\log q$ into a universal singular skeleton and a smooth part:

Definition 2 (The LQD slice)**.** Fix an order $N \geq 4$ and coefficients $\theta = (L, R, a_2, \ldots, a_N)$. The LQD quantile density is

$$\boxed{q(u) = \frac{e^{g(u)}}{u\,(1-u)}, \qquad g(u) = (1-u)\,L + u\,R + \sum_{n=2}^{N} a_n\,P_n(1-2u),} \tag{5}$$

for $u \in (0,1)$, with P_n the Legendre polynomials. The quantile is $Q(u) = \mu + \int_{1/2}^{u} q$, with the scalar μ fixed by the martingale condition $\mathbb{E}[e^X] = \int_0^1 e^{Q(u)} \mathrm{d}u = 1$.

Equivalently $\log q(u) = -\log u - \log(1-u) + g(u)$: the two logarithmic terms are the skeleton, the smooth g is everything else. The shipped default $N = 6$ carries seven parameters; the API accepts $4 \le N \le 16$, and higher N adds body detail without ever touching the no-arbitrage mechanism. Inverting $q = 1/f_X(Q)$ gives the density at the rank u:

$$f_X(Q(u)) = \frac{1}{q(u)} = u\,(1-u)\,e^{-g(u)} \;\ge\; 0 \qquad \text{for every } \theta, \tag{6}$$

and this single line is the whole no-arbitrage mechanism. In code:

Listing 1: The LQD density is structurally non-negative — no constraint, no penalty (distilled from `models/lqd/quadrature.py`).

```
def density(u, g):           # g(u) = (1-u) L + u R + sum a_n P_n(1-2u)
    return u * (1 - u) * np.exp(-g)    # >= 0 for ANY g: no butterfly
```

Proposition 2 (A single LQD slice is butterfly-free)**.** *Suppose $q > 0$ on $(0,1)$ and the martingale integral $\int_0^1 e^{Q(u)}\mathrm{d}u$ is finite. After the scalar shift of Q that makes this integral one, the call curve $C(k) = \int_{u_k}^1 (e^{Q(u)} - e^k)\,\mathrm{d}u$, with u_k the rank where $Q(u_k) = k$, is generated by a mean-one probability law on $Y > 0$: the slice is free of butterfly arbitrage, and put–call parity $C(k) - P(k) = 1 - e^k$ holds identically.*

Proof. Since $q > 0$, Q is strictly increasing and is the quantile of a genuine law for X; then $Y = e^X > 0$ and the shift enforces $\mathbb{E}[Y] = 1$. The displayed $C(k)$ is just $\mathbb{E}[(Y - e^k)^+]$ written rank-by-rank (u is a uniform variable and $Y = e^{Q(u)}$), so monotonicity and convexity in strike follow from (4); the put integral $P(k) = \int_0^{u_k} (e^k - e^{Q(u)})\,\mathrm{d}u$ subtracts to $C - P = \int_0^1 (e^Q - e^k) = 1 - e^k$. No positivity inequality is ever checked. □

Heuristic

Why does the quantile side succeed where the density side struggled? Because exponentiating kills a *pointwise* constraint but not an *integral* one. On the density side, $f = e^{\phi}$ buys positivity but leaves two integral constraints ($\int f = 1$, $\int e^x f = 1$) coupling all parameters nonlinearly. On the quantile side, $q = e^{\ell}$ buys monotonicity, the mass constraint is *automatic* (every increasing Q on $(0,1)$ is a quantile — ranks are born normalized), and the one remaining integral constraint is solved, not imposed: shifting Q by a constant multiplies $\mathbb{E}[e^X]$ by e^{μ}, so one scalar μ absorbs it exactly, leaving the shape of the law untouched. One chart, zero pointwise constraints, zero residual equality constraints.

Why this is the right trade. A direct-volatility model spends optimizer effort policing arbitrage; the LQD model spends none, because the feasible set *is* the set of laws. The price is two scalar nonlinearities — the shift μ (one logarithm of one integral, section 7) and the tail admissibility $A_R < 1$ (section 4) — both imposed once per residual evaluation, not at every quote.

4 Tails first: the endpoint skeleton

A basis for g should be chosen tails-first, because extrapolation is where smile models die. The far wings of a slice are constrained by a classical and completely model-free result, Lee's moment formula [2], and the two logarithmic terms in (5) are engineered so that LQD lands *inside* Lee's bounds automatically, with the wing slopes carried by two readable parameters. This section owns that story; the cross-model treatment of wings is Note 09's.

4.1 Endpoint scales are quantile slopes

The skeleton terms diverge at the ranks' endpoints, and the smooth part g decides *how fast.* Because $P_n(1) = 1$ and $P_n(-1) = (-1)^n$, the two endpoint values of g are

$$A_L := e^{g(0)} = \exp\Big(L + \sum_{n\geq 2} a_n\Big), \qquad A_R := e^{g(1)} = \exp\Big(R + \sum_{n\geq 2}(-1)^n a_n\Big). \tag{7}$$

Their meaning is best read in the log-odds coordinate $z = \log\frac{u}{1-u}$ that will run the whole pricing machine. Since (writing $u(z)$ for the inverse map $u = 1/(1+e^{-z})$, whose derivative is $u(1-u)$)

$$\frac{\mathrm{d}Q}{\mathrm{d}z} = q(u)\,u(1-u) = e^{g(u(z))}, \tag{8}$$

the quantile has *finite slope* e^g in z everywhere, and at the two ends

$$Q(z) = \mu + A_L\,z + O(1)\ \ (z\to -\infty), \qquad Q(z) = \mu + A_R\,z + O(1)\ \ (z\to +\infty). \tag{9}$$

So A_L and A_R are the asymptotic speeds at which log-return quantiles recede per unit of log-odds: genuine tail *scales*, not diagnostics. Read (7) once more with a fitter's eye, though: *every* body coefficient a_n enters one or both sums, so in the raw coordinates (L, R, a) the tails and the body are coupled — L and R alone are *not* readable tail handles. Section 5.1 makes $(\log A_L, \log A_R)$ themselves the coordinates, which is where every "endpoint terms own the tails" statement in this lecture is meant to be read.

4.2 Exponential tails, the moment strip, and the one hard wall

Linear quantiles in log-odds mean exponential tails in return space. To see it, ask how much probability lives above the return level $x = Q(z)$: exactly the rank remainder $1 - u(z)$, which behaves like e^{-z} for large z. Inverting the asymptote $x \approx \mu + A_R z$ gives $z \approx (x-\mu)/A_R$, so

$$\mathbb{P}(X > x) \asymp e^{-x/A_R} \quad (x\to+\infty), \qquad \mathbb{P}(X < x) \asymp e^{x/A_L} \quad (x\to-\infty): \tag{10}$$

each tail is exponential, with decay rate the reciprocal of its scale.

Moments now follow from a single change of variables, and it is worth doing once slowly because the whole admissibility story drops out. Any expectation is a rank integral, and in log-odds the rank measure carries an intrinsic tent-shaped weight:

$$\mathbb{E}[h(X)] = \int_0^1 h(Q(u))\,\mathrm{d}u = \int_{-\infty}^{\infty} h(Q(z))\,\underbrace{u(z)(1-u(z))}_{\asymp e^{-|z|}}\,\mathrm{d}z. \tag{11}$$

Take $h(x) = e^{rx}$. On the right, $e^{rQ(z)} \asymp e^{rA_R z}$ against the weight e^{-z}: the integrand behaves like $e^{(rA_R-1)z}$, integrable at $+\infty$ iff $rA_R < 1$. On the left, $e^{rQ(z)} \asymp e^{rA_L z}$ against the weight e^{+z}: the integrand behaves like $e^{(rA_L+1)z}$, integrable at $-\infty$ iff $rA_L > -1$. So $\mathbb{E}[e^{rX}]$ converges precisely on the strip

$$-\frac{1}{A_L} < r < \frac{1}{A_R}\,, \tag{12}$$

each endpoint lost exactly when the exponential growth of e^{rQ} ties the exponential decay of the weight. In return moments — recalling $Y = e^X$ — the strip says that the last finite moments are

$$p_R^* = \sup\{p \geq 0 : \mathbb{E}[Y^{1+p}] < \infty\} = \frac{1}{A_R} - 1, \qquad p_L^* = \sup\{p \geq 0 : \mathbb{E}[Y^{-p}] < \infty\} = \frac{1}{A_L}. \tag{13}$$

One point of the strip is load-bearing. The martingale integral $\mathbb{E}[e^X]$ is the case $r = 1$, and $r = 1$ lies inside the strip iff

$$A_R < 1 \qquad \text{(finite forward)} \tag{14}$$

— the single wall promised in table 2. No left-hand condition exists because $r = 1 > -1/A_L$ always. The wall is not an artefact of the basis: any law whose right return tail is exponential with scale ≥ 1 has an infinite mean, whatever model writes it down.

Caution

$A_R < 1$ is a genuine hard constraint, not a regularization preference: a parameter vector with $A_R \geq 1$ assigns the forward an *infinite* mean, and its "prices" are meaningless. The calibrator enforces it twice — a hard rejection at $A_R \geq 1 - \varepsilon_R$ and a smooth softplus barrier that engages well before the wall (section 9) — so both the analytic and the finite-difference Jacobians stay smooth as the optimizer approaches the boundary. Fits that live near the wall should be treated as fragile even when admissible (section 14).

4.3 From moments to Lee slopes

Lee's moment formula turns (13) into wing geometry. Writing $\beta_R = \limsup_{k\to+\infty} w(k)/k$ and $\beta_L = \limsup_{k\to-\infty} w(k)/|k|$ for the asymptotic slopes of total implied variance, Lee proved

$$\beta = \psi(p), \qquad \psi(p) = 2 - 4\big(\sqrt{p^2 + p} - p\big) \in (0, 2], \tag{15}$$

with p the relevant critical exponent from (13). For LQD,

$$\beta_L = \psi\Big(\frac{1}{A_L}\Big), \qquad \beta_R = \psi\Big(\frac{1}{A_R} - 1\Big), \qquad A_R < 1. \tag{16}$$

The endpoint coefficients therefore do double duty: they regularize extrapolation *and* identify the moment budget and the Lee-consistent wing slopes. A calibration may fit L, R freely (subject only to the wall) or impose explicit wing priors by pinning A_L, A_R through (7). Figure 3 draws the two maps.

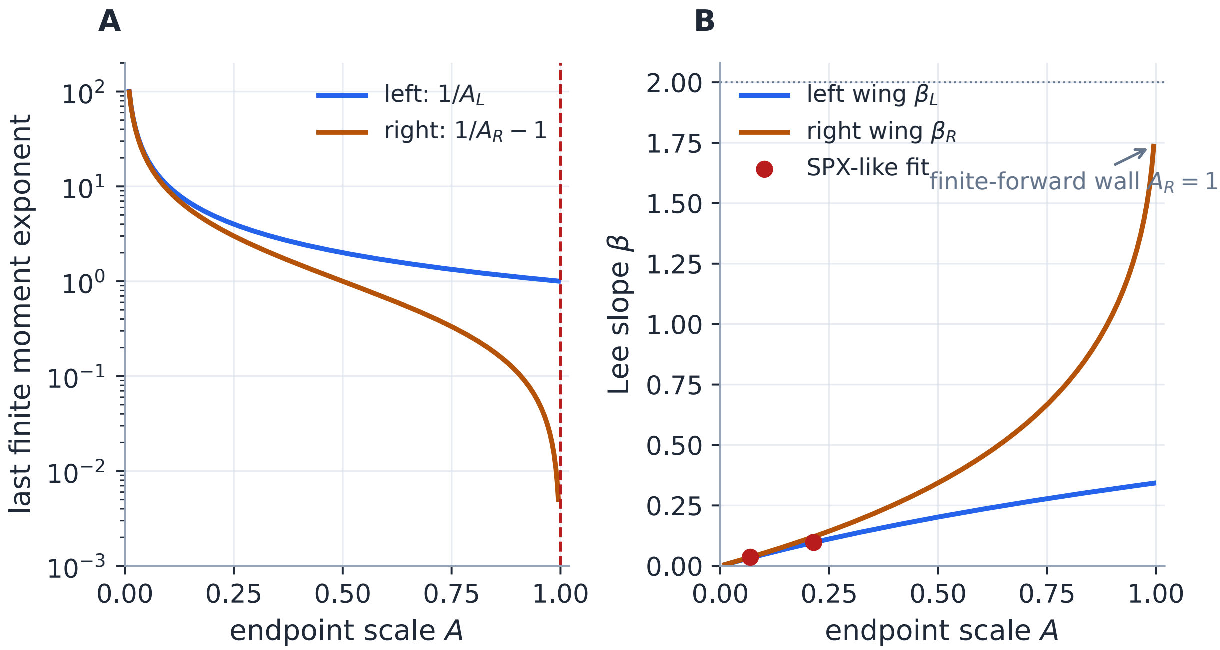

Figure 3: The tail dial. Panel A: the last finite moment exponent as a function of the endpoint scale, $1/A_L$ on the left tail and $1/A_R - 1$ on the right; the dashed line is the finite-forward wall $A_R = 1$, where the moment budget of the right tail is exhausted. The wall belongs to the RIGHT scale only: A_L has no upper bound — the shared axis range is a drawing convenience, not a constraint. Panel B: the Lee map ψ carries the same scales to asymptotic wing slopes, capped by the model-free ceiling $\beta = 2$; the dots mark an SPX-like production fit, sitting far below the ceiling — equity smiles use only a sliver of the admissible range.

Heuristic

The map ψ is increasing in the tail scale: a *heavier* tail (larger A, smaller critical exponent) makes a *steeper* variance wing. Be precise about "fat": LQD return tails are *exponential* — asymptotically far heavier than the log-normal's Gaussian log-return tails, though over traded strikes the two can look close. Equity index fits land at small A_L, A_R (our running example fits $A_L = 0.157473$, $A_R = 0.037786$), so their wings sit comfortably inside Lee's bounds, and truncating the log-odds line at $|z| \le 40$ costs nothing at double precision (section 7).

5 The Legendre body

With the tails owned by the skeleton, the body of g wants a basis that is smooth, global, well-conditioned, and orthogonal — deviations along different basis directions should not masquerade as one another. On $x = 1 - 2u \in (-1, 1)$ the Legendre polynomials satisfy

$$\int_0^1 P_m(1-2u)\, P_n(1-2u)\,\mathrm{d}u = \frac{\mathbf{1}_{m=n}}{2n+1}, \tag{17}$$

and are generated by the stable three-term recursion

$$(n+1)P_{n+1}(x) = (2n+1)\,x\,P_n(x) - n\,P_{n-1}(x), \qquad P_0 = 1,\ P_1 = x. \tag{18}$$

The expansion in (5) starts at $n = 2$ deliberately: degrees zero and one are already spanned by the endpoint terms, since

$$(1-u)L + uR = \frac{L+R}{2} + \frac{L-R}{2}(1-2u). \tag{19}$$

So the $N = 6$ model spans exactly the polynomial space of $P_0, \ldots, P_6$ while keeping the first two coordinates meaningful as *tail* handles rather than abstract polynomial coefficients: $\frac{L+R}{2}$ is an overall scale (roughly, the level of volatility) and $\frac{L-R}{2}$ a tail asymmetry (roughly, the skew).

What does a body mode actually *do* to a distribution? The mechanism is worth internalizing, because it is how every LQD fit works. Raising g near some rank u stretches the quantile spacing there ($q = e^g/u(1-u)$ grows), and stretched spacing is *thinned probability*: the density $1/q$ falls on that stretch of returns while the displaced mass reappears wherever g was lowered. A body mode is therefore a mass-shuffling instruction written in rank coordinates, and its symmetry decides its trading meaning. $P_n(1-2u)$ is symmetric under $u \leftrightarrow 1-u$ for even n and antisymmetric for odd n, so *even modes make symmetric reshapes, odd modes make skew-like ones.* Concretely, for the first three (shown in fig. 4 on a symmetric reference):

- $a_2 > 0$: P_2 is $+1$ at both endpoints and $-\frac{1}{2}$ at the median, so both tails stretch while the centre compresses — a sharper peak, fatter tails, a more convex smile. This is the butterfly mode.
- $a_3 > 0$: P_3 is antisymmetric, raising one tail's g and lowering the other's — mass slides from one side to the other, and the smile tilts. This is the risk-reversal mode.
- a_4: one more oscillation of the same game, sculpting the *shoulders* between body and tail.

Higher modes continue the pattern with finer brushes. And because fig. 4 shows the same three perturbations in three charts at once, it is also a small test of the reader's new fluency: cover two panels and predict the third.

One caveat belongs in the same breath, because it is the section's honest half. $P_n(1) = 1$ and $P_n(-1) = (-1)^n$, so every raw body coefficient also moves the endpoint *values* of g — by (7), $a_2 \mapsto a_2 + 0.10$ multiplies *both* tail scales by $e^{0.10} \approx 1.105$, and the production audit of fig. 5 confirms the factor to four digits. In raw coordinates, fitting a shoulder quote can therefore drag the last finite moment and the ultimate wing slope as a side effect. This is not a defect of the model — the fitted *smile* is coordinate-independent — but it is a defect of the raw coordinates as a language for priors, manual edits and warm starts, and the next subsection removes it exactly.

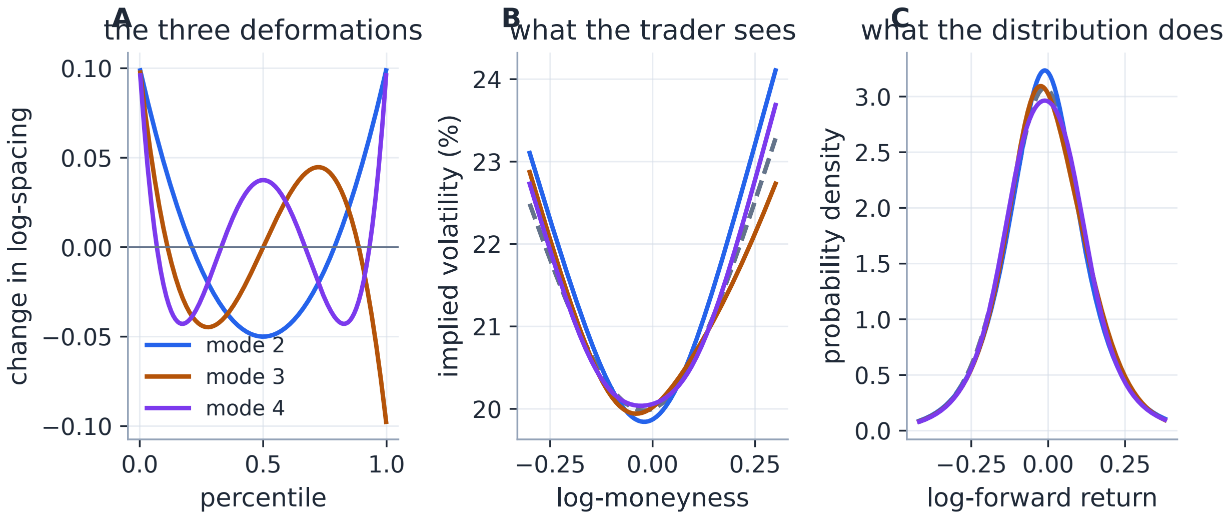

Figure 4: One production basis coefficient at a time: a_2, a_3 or a_4 set to 0.10 on the symmetric logistic reference of section 6, everything else untouched. Panel A: the perturbation of g — a pure Legendre mode in the rank, even ones symmetric, odd ones not. Panel B: what the trader sees — mode 2 bends the smile symmetrically (butterfly), mode 3 tilts it (risk reversal), mode 4 works the shoulders. Panel C: what the distribution does — every rank interval keeps its probability, but the rank-to-return map stretches where g rises, so the density thins there and thickens where g falls; every deformed density is non-negative without anyone checking. The moral — *body modes shuffle mass, endpoint values own the tails* — is exactly true only in the decoupled chart of section 5.1, because each raw mode drawn here also multiplies both tail scales by $e^{0.10}$.

5.1 The endpoint chart, and the coordinate that removes the wall

The cure for the coupling is not a new model; it is the model's own coordinates, chosen so the quantities with names sit on the axes. Take as coordinates the two endpoint *values* and the body:

$$\phi = \big(\log A_L,\ \log A_R,\ a_2, \ldots, a_N\big), \qquad \theta = M\phi, \tag{20}$$

where M is the exact, unit-determinant linear map that back-solves (L, R) from (7) — holding (ϕ_0, ϕ_1) fixed while moving a body coordinate counter-adjusts (L, R) to keep both tail scales put. Equivalently, the body basis functions become

$$\varphi_n(u) = P_n(1 - 2u) - (1 - u) - (-1)^n u, \qquad \varphi_n(0) = \varphi_n(1) = 0: \tag{21}$$

endpoint-vanishing modes spanning the *same* polynomial space, so this is a change of basis, not of family — same slices, same optimum, different names on the axes. Figure 5 shows the difference in one experiment on the fitted running example.

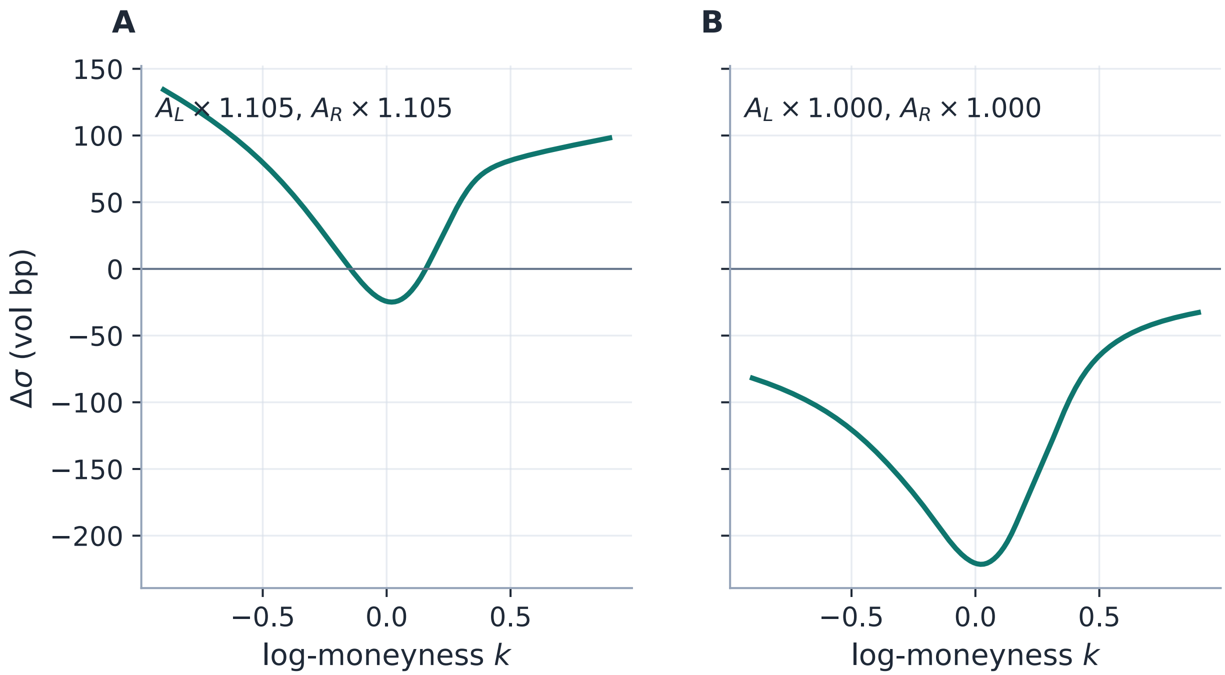

Figure 5: The same $+0.10$ on the "butterfly" coordinate, in two charts, applied to the fitted running example. Panel A: raw a_2 — the smile bends, and both tail scales are dragged by exactly $e^{0.10}$ ($A_L \times 1.105$, $A_R \times 1.105$, the production numbers printed in the panel). Panel B: the endpoint-chart body coordinate — the belly moves, and both tail scales stay at factor 1.000: the wings are pinned *by construction*, not by luck. Shoulder fitting, wing priors and manual edits speak different coordinates well only in the second chart.

One coordinate remains half-named: $\log A_R$ must stay below zero. Push it through the logistic,

$$A_R = \operatorname{expit}(\rho) = \frac{1}{1 + e^{-\rho}}, \qquad \rho \in \mathbb{R}, \tag{22}$$

and the wall is not merely respected — it is *unreachable*: every $\psi = (\log A_L, \rho, a_2, \ldots, a_N) \in \mathbb{R}^d$ maps to an admissible slice, the chart covers exactly the admissible set, and the title's word "unconstrained" becomes a literal statement about the production solve. Near the wall the chart compresses ($\mathrm{d}\log A_R/\mathrm{d}\rho = 1 - A_R \to 0$), damping trust-region steps precisely where the raw charts needed a barrier; far below it, $\rho \approx \log A_R$ and nothing changes. The objective is identical in every chart, so the optimum is chart-independent up to solver tolerance: refitting the running example in the raw and logistic charts agrees to 2.6×10^{-8} in the coefficients and 4.2×10^{-6} vol basis points in fit quality, and a twelve-node live-market validation (section 15) reproduces the same agreement on real chains.

Remark 1. The chart changes the *language*, not the *information*. Tail scales decoupled from body modes are still tail scales identified only by wing quotes, and section 12.1 measures how weakly; the barrier of section 9 also remains part of the objective — an economic prior against near-infinite forwards, no longer a feasibility device. Two practical dividends are worth naming: warm starts expressed in ψ survive re-ordering of the body (the tails do not lurch when N changes), and floating-point failure at the old wall is replaced by an explicit overflow budget (section 7.2) —

cancellation between huge (L, R) and huge body coefficients can hide an interior blow-up of g from any endpoint test, so the guard watches g itself, not the wall.

6 An exactly solvable slice

Every lecture needs its harmonic oscillator. Set all body modes to zero and the two endpoint coefficients equal:

$$a_n = 0, \qquad L = R = \log s \qquad \Longrightarrow \qquad g \equiv \log s, \quad \frac{\mathrm{d}Q}{\mathrm{d}z} = s. \tag{23}$$

The quantile is exactly linear in log-odds, $Q(z) = \mu + s\,z$; that is, X is a *logistic* random variable with scale s: $X = \mu + sZ$ with $Z = \log \frac{U}{1-U}$ for a uniform rank U.

Everything about this slice flows from one integral. The moment generating function of Z is a rank integral we can actually do — substituting $e^{tZ} = \left(\frac{u}{1-u}\right)^t$ turns it into Euler's beta integral:

$$\mathbb{E}\left[e^{tZ}\right] = \int_0^1 \left(\frac{u}{1-u}\right)^t \mathrm{d}u = \mathrm{B}(1+t,\, 1-t) = \Gamma(1+t)\,\Gamma(1-t) = \frac{\pi t}{\sin(\pi t)}, \qquad |t| < 1. \tag{24}$$

Note, in passing, that the formula itself confirms section 4: it exists exactly on the strip $|t| < 1$, as it must for a law with $A_L = A_R = 1$ per unit scale. Two facts fall out of (24) at once:

- **The martingale shift, exactly.** Read it at $t = s$: $\mathbb{E}[e^{sZ}] = \pi s/\sin(\pi s)$, so the shift that makes e^X mean-one is $\mu = -\log\left(\pi s/\sin(\pi s)\right)$ in closed form — the only slice in the family whose μ needs no quadrature.

- **The variance, and hence the initializer.** Expand its logarithm at $t = 0$: since $\sin(\pi t) = \pi t\,(1 - \pi^2 t^2/6 + \cdots)$,

$$\log \mathbb{E}\left[e^{tZ}\right] = \frac{\pi^2 t^2}{6} + O(t^4) \qquad \Longrightarrow \qquad \mathrm{Var}(Z) = 2 \cdot \frac{\pi^2}{6} = \frac{\pi^2}{3},$$

  the second cumulant being twice the quadratic coefficient. So $\mathrm{Var}(X) = \pi^2 s^2/3$, and matching a target ATM total variance w_0 gives the seed $s \approx \sqrt{3w_0}/\pi$.

The remaining diagnostics are immediate: $A_L = A_R = s$, so the slice is admissible iff $s < 1$ (a 55% vol at one year is $s \approx 0.30$, nowhere near the wall), and the Lee slopes are $\beta_L = \psi(1/s)$ and $\beta_R = \psi(1/s - 1)$ — slightly asymmetric wings even for this symmetric law, because the martingale shift tilts the smile.

This two-line model is precisely the production cold-start initializer — the `logistic_init` routine of `models/lqd/calibrate.py`: match s to the ATM-interpolated quote variance, then let the optimizer bend $L \neq R$ (tail asymmetry) and the a_n (body).

Figure 6 walks the whole pipeline on a solved instance — production root-solving the scale so that the ATM volatility is *exactly* 20% at $\tau = 0.50$ — and it repays reading one panel at a time. Panel A is the coordinate itself: the log-odds line holds every percentile of an unbounded law at finite coordinates, with equal steps in z meaning equal *multiplications* of the odds (the 1st and 99th percentiles sit at $z = \mp 4.6$, not at infinity). Panel B is the law: in rank coordinates the straight line $Q = \mu + sz$ becomes the gentle S-curve shown, and one detail deserves a pause — the forward ($k = 0$, marked) sits at the 53.3436% rank, *above* the median. That is the martingale shift at work:

for e^X to average to one, the convexity of the exponential demands a median return slightly below zero ($Q(\frac{1}{2}) = \mu = -0.01088288$), so more than half the outcomes finish below the forward. Jensen's inequality, drawn. Panel C is the market's view of the same object: a nearly symmetric, gently convex smile pinned at 20.00% — *nearly*, because the symmetric law does not make a symmetric smile: the martingale shift tilts it, which is the same fact the unequal Lee slopes $\beta_L \neq \beta_R$ stated a paragraph ago. Convex, because exponential tails are heavier than the log-normal's, so far strikes carry more relative value, which the Black formula can only express as higher implied volatility. The example box then prices one option on this slice by hand.

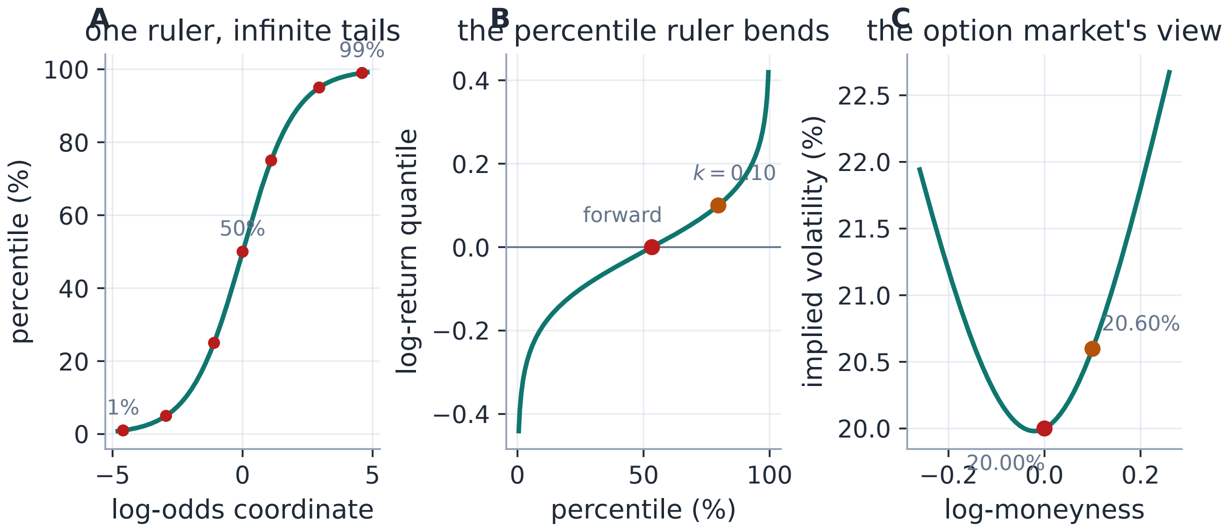

Figure 6: The exactly solvable slice, solved to an ATM volatility of exactly 20% at $\tau = 0.50$ (production root-solve: $s = 0.08125025$). A: the log-odds ruler — every percentile at a finite coordinate. B: the quantile in rank coordinates; the forward's rank is 53.3436% > 50% because the martingale shift pushes the median return down to $\mu = -0.01088288$ (Jensen's inequality, drawn). C: the resulting smile — 20.00% at the money by construction, convex because exponential tails out-fatten the log-normal.

A trade ticket priced by hand

On the solved slice of fig. 6, price the $k = 0.10$ call ($\approx 10.5\%$ out of the money). The strike's rank is $u_k = 0.796524$: the option finishes in the money with probability $1 - u_k \approx 20.3\%$. The two ledgers of section 7 read $G(z_k) = 0.24722094$ (the share of the mean carried by outcomes above the strike) and $e^k(1 - u_k) = 0.22487593$ (the strike paid on that event), so

$$C(0.10) = G - e^k(1 - u_k) = 0.02234501 \qquad \Longrightarrow \qquad \sigma_{\text{imp}} = 20.5977\%,$$

about 0.6 vol points above the ATM level: the first taste of a smile. Production reproduces every number in this box to the displayed digits (`gen_lqd_fresh.py`); the martingale check on the slice returns $\mathbb{E}[e^X] - 1 = 0$ to machine precision.

7 From coefficients to prices: one cumulative integral

The whole pricing pipeline is the arrow chain

$$\theta \longrightarrow g \longrightarrow Q \longrightarrow \mu \longrightarrow C(k) \longrightarrow \sigma(k),$$

and this section makes each arrow one integral (or one root-solve) on a single shared grid.

7.1 The log-odds machine

The skeleton's endpoint singularities are a feature for tails and a nuisance for quadrature — so we integrate in the coordinate that removes them. Under $z = \log \frac{u}{1-u}$, by (8), the quantile is the integral of a smooth positive function on the whole line:

$$Q(z) = \mu + \int_0^z e^{g(u(s))} \, \mathrm{d}s, \tag{25}$$

anchored at the median node $z = 0$. The shift is fixed by the martingale condition, which in the same coordinate reads (using $\mathrm{d}u = u(1-u)\,\mathrm{d}z$)

$$\mu = -\log\left(\int_{-\infty}^{\infty} e^{\,Q(z)-\mu}\, u(z)\big(1-u(z)\big)\,\mathrm{d}z\right), \tag{26}$$

one logarithm of one integral. This is the *only* normalization in the model; being an additive shift of Q it changes neither q, nor the tails, nor positivity — it slides the whole law along the return axis until the forward is its mean.

Pricing needs one more cumulative object. Define the *upper share*

$$G(z) = \int_z^{\infty} e^{\,Q(s)}\, u(s)\big(1-u(s)\big)\,\mathrm{d}s \;=\; \mathbb{E}\big[\,Y\,\mathbf{1}_{\{Y>e^{Q(z)}\}}\big] : \tag{27}$$

the share of the (unit) mean carried by outcomes above the rank $u(z)$. It falls from $G(-\infty) = 1$ to $G(+\infty) = 0$, and it prices every call at once: if z_k solves $Q(z_k) = k$, then, splitting the call payoff into asset and cash legs,

$$C(k) = G(z_k) - e^k\,\big(1 - u(z_k)\big). \tag{28}$$

The put is $P(k) = e^k u(z_k) - (1 - G(z_k))$, and parity is immediate. Equation (28) is worth staring at: an option price is the difference of two ledgers, both monotone, both already on the grid. The same G will hand us the analytic Jacobian in section 10 and the calendar ordering in section 11 — one object, three jobs.

7.2 What one build of a slice actually is

It is worth being concrete here, because the entire calibrator stands on this one loop being cheap and smooth. Given a trial θ, production does five things, in order, on a fixed symmetric grid $z_j \in [-Z, Z]$:

1. evaluate g at the (cached) ranks u_j — one matrix-vector product, since g is linear in θ;

2. integrate e^g cumulatively (fourth-order Simpson) to get the unshifted quantile, anchored at the median node $z = 0$;

3. compute the martingale integral, take one logarithm, and shift: $Q \mathrel{+}= \mu$;
4. integrate $e^Q u(1-u)$ cumulatively *from the right* to get the upper share G;
5. stop. Every strike, at any later moment, is priced from (28) by interpolating these two curves and solving one root.

Nothing here is iterative, and the cost is a few dense passes over one array. "Nothing can fail" would be one claim too many: floating point can, in exactly two ways, and both are budgeted rather than hoped away. An interior excursion of g can overflow e^g or the running integral even while both endpoint values stay admissible (endpoint cancellation hides a body blow-up from any wall test), so the build refuses, with an explicit exponent budget, any trial whose g or unnormalized quantile exceeds 700 — a clean rejection the calibrator's penalty branch absorbs, never an `inf` in a price. And at $|z| \gtrsim 36.7$ double precision rounds $u = \operatorname{expit}(z)$ to exactly one, so every $1-u$ and $u(1-u)$ in the pipeline is evaluated on its own accurate side ($u(1-u) = \operatorname{expit}(z)\operatorname{expit}(-z)$; products with e^k in log space) — section 7.4 reports what the audit that found this was worth.

Two details of this recipe deserve their "why". The first is the truncation at $|z| = Z$ (production: $Z = 40$). Beyond the grid the integrands of steps 3 and 4 are, by section 4, essentially exact exponentials — so instead of praying the tail is negligible, production *sums it in closed form*, appending the geometric-series corrections

$$\int_Z^{\infty} e^Q u(1-u)\,\mathrm{d}z \approx \frac{e^{Q(Z)-Z}}{1-A_R}, \qquad \int_{-\infty}^{-Z} e^Q u(1-u)\,\mathrm{d}z \approx \frac{e^{Q(-Z)-Z}}{1+A_L}. \tag{29}$$

(Notice the right-hand denominator displaying the $A_R = 1$ wall once more.) For equity tail scales the corrections themselves are far below double-precision noise at $Z = 40$; they are kept because they make the build exact-in-the-limit rather than merely truncated.

The second detail is how the two curves are read *between* nodes. The construction hands us the exact nodal derivatives for free — $\mathrm{d}Q/\mathrm{d}z = e^g$ from (8) and $\mathrm{d}G/\mathrm{d}z = -e^Q u(1-u)$ from (27) — and it would be wasteful not to use them: production stores both and interpolates off-grid with *cubic Hermite* polynomials rather than linearly. The payoff is threefold: priced curves are smooth enough for clean finite-difference Greeks; the strike root $Q(z_k) = k$ can be polished by Newton on the interpolant to machine precision (which is what will make the ATM handles of section 8 *exact* rather than grid-limited); and because Hermite interpolation returns the nodal value exactly at a node, a constraint expressed at grid z-values — the calendar floor of section 11 — matches its node-indexed form bit for bit, whatever grid the optimizer runs on.

Performance

The grid $(z_j, u_j, u_j(1-u_j))$ and the Legendre matrix $P_n(1-2u_j)$ depend only on (Z, n_{pts}, N) — never on θ. A calibration builds the slice many hundreds of times, so these arrays are LRU-cached and returned read-only; only the parameter combination g is re-formed per call, which alone made `build_slice` about 1.7× faster. Optimization iterates on a coarser grid of 2001 nodes (Simpson error already orders of magnitude below the fit budget) and the accepted slice is rebuilt once at the full 8001 nodes for reporting, density, var-swap and calendar diagnostics. Converged parameters agree with a full-grid fit to $\sim 10^{-6}$. See appendix B.

7.3 Back to the quoted smile

The market speaks implied volatility, so the last arrow inverts Black. In normalized total-variance units,

$$B(k, w) = \Phi(d_+) - e^k \, \Phi(d_-), \qquad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}, \tag{30}$$

and the model's total implied variance solves $B(k, w(k)) = C(k)$ — a scalar root per strike, monotone in w, safe for Newton.

Calibration, however, deliberately does *not* fit in volatility space. Given a quote σ_i at k_i, with $w_i = \sigma_i^2 \tau$, the residual is the *vega-normalized price error*

$$r_i(\theta) = \frac{C(k_i; \theta) - B(k_i, w_i)}{\partial_\sigma B(k_i, w_i) + \eta}, \qquad \partial_\sigma B = \varphi(d_+)\sqrt{\tau}, \tag{31}$$

with a small floor η against the vanishing far-wing vega (appendix A). Working in price space keeps every iterate a valid law (no implied-vol inversion inside the loop, no inversion failures on wild iterates); dividing by vega undoes the price-to-vol scale so the optimizer still sees, to first order, a volatility error. The error *ledger* the note reports — always in vol basis points — is computed after convergence by honest Black inversion.

The floor η is not innocent, and deserves its own figure. Wherever vega falls below η the residual stops being a volatility error and becomes a (scaled) price error: the *effective* weighting regime is strike- and maturity-dependent. Figure 7 draws it: at $\tau = 0.75$ the floor never binds across the plotted strikes; at a month it binds beyond $|k| \approx 0.25$; at two days, beyond $|k| \approx 0.07$ — short-dated wings are fitted under a flat price-error ceiling, not a vol metric. This is a deliberate trade (a vol metric at zero vega would amplify noise without bound), but a reader of short-dated wing fits should know which regime their quotes sat in.

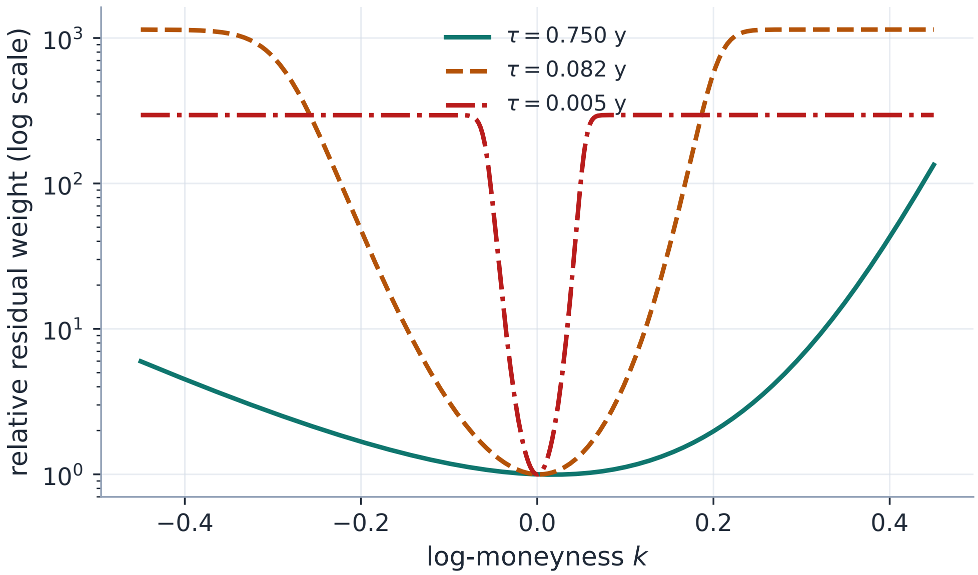

Figure 7: The vega floor's effective weighting regime. Relative residual weight $1/(\partial_\sigma B + \eta)$, normalized at the money, on the running example's smile shape rescaled to three maturities. The plateaus are where the floor binds: there the objective weighs quotes by price, not volatility, and the plateau height is the ratio of ATM vega to the floor. The crossover strike moves inward as maturity shortens — the regime is a property of (k, τ) jointly, which is why it is drawn, not asserted.

7.4 What the implementation certifies

Proposition 2 is a statement about the *continuous* law. Production prices through interpolants of Q and G, and an interpolant inherits nothing automatically: ordinary cubic Hermite pieces are not monotonicity-preserving in general (positive nodal derivatives do not preclude segment overshoot), and between nodes the two curves are no longer algebraically consistent. The honest guarantee is therefore two-part: *a structural theorem for the continuous object, plus a certificate and a measured tolerance for the implementation.* Both parts are cheap, and both are checked, not trusted.

The certificate. For a cubic Hermite interpolant of an increasing function on a uniform grid, the classical Fritsch–Carlson condition is sufficient: on each segment, monotonicity holds if both endpoint derivatives lie in $[0, 3 \times \text{secant slope}]$. With *exact* nodal derivatives of a smooth curve the derivative-to-secant ratio is $1 + O(h^2)$ — far inside the region — so the condition is verifiable per slice as a vectorized scan, and it upgrades "increasing at the nodes" to "increasing *between* them, proven" (with one honest footnote: segments where the asset share has decayed below round-off are certified flat-to-tolerance rather than strictly monotone).

The audit. What the certificate does not cover — convexity of the priced composition, butterflies at finite width, digital bounds — a randomized audit measures. Sixty parameter vectors drawn *through the logistic chart* (so all admissible by construction), spanning orders 4–16, plain, near-wall (A_R up to 0.993) and wild-body classes, are priced at thousands of off-grid strikes and checked in *strike* space — convexity lives in $K = e^k$, not in k, where a deep-ITM call is legitimately concave. The worst violations across the whole battery: call-price bounds 2.5×10^{-9}, butterflies (four widths) 3.0×10^{-13}, digitals 3.7×10^{-12}, and agreement with a $4\times$ finer reference quadrature at random sub-grid strikes 2.9×10^{-9} — one tolerance set, *uniform across the classes*, quotable as: the implementation's residual arbitrage is bounded by about 10^{-9} of the forward.

What the audit caught. Run first against an earlier build, the same battery failed — and the failure is worth a paragraph precisely because a referee predicted its species. At $|z| \gtrsim 36.7$, $\operatorname{expit}(z)$ rounds to exactly 1, so the pricing identity's $e^k(1-u_k)$ leg silently collapsed to zero and the far right wing stepped up to $G(z_k)$ — immaterial for equity-like tails, but near the wall those strikes still carry prices of order 10^{-1}. The fix is log-space evaluation throughout (section 7.2); the audit's before and after is the difference between a 10^{-2} violation and 3.0×10^{-13}. Strikes beyond the grid's quantile range are priced on the slice's own tail asymptote,

$$C(k) = e^{\,k - z_R(k)}\,\frac{A_R}{1 - A_R}, \qquad z_R(k) = Z + \frac{k - Q(Z)}{A_R}, \tag{32}$$

(mirrored for far-left puts), continuous at the seam with (29) and Lee-consistent by construction — the display edge of a short-dated smile genuinely reaches this region. *A model whose guarantees are structural can still leak at the last bit; the audit is what turns "cannot have arbitrage" into "measured not to, at tolerance so-and-so".*

8 Three exact handles, and a chart built around them

8.1 Level, skew, curvature in closed form

Traders want the leading coordinates of a smile model to be the *actual* ATM volatility, skew and curvature — not proxies, not finite differences of a plotted curve. The LQD slice delivers all three exactly, and the derivation is three short steps: read two numbers off the law, pin the ATM level, then push everything through the Black formula with the chain rule. Recall the convention of table 1: primes are k-derivatives of one-variable curves (C, w, σ), subscripted ∂'s are partials of the two-variable $B(k, w)$; everything below is evaluated at the money.

Step 1: two numbers from the law. Differentiate the rank-form call $C(k) = \int_{u_k}^{1} (e^{Q(u)} - e^k)\,\mathrm{d}u$ in k. The moving boundary u_k contributes nothing — the integrand vanishes exactly there, an envelope fact that returns in force in section 10 — so only the $-e^k$ inside survives: $C'(k) = -e^k(1-u_k)$. Differentiating once more brings in the density, since $\mathrm{d}u_k/\mathrm{d}k = f_X(k)$. At $k = 0$, writing u_0 for the forward's rank ($Q(u_0) = 0$) and $f_0 = u_0(1-u_0)e^{-g(u_0)}$ for the ATM density (6),

$$C'(0) = -(1-u_0), \qquad C''(0) = f_0 - (1-u_0). \tag{33}$$

Step 2: the ATM level. At $k = 0$ the Black formula collapses, and the ATM total variance w_0 is pinned by one monotone scalar equation,

$$B(0, w_0) = 2\,\Phi\Big(\frac{\sqrt{w_0}}{2}\Big) - 1 = C(0). \tag{34}$$

Step 3: the chain rule through Black. The model's smile is *defined* by the identity $B\big(k, w(k)\big) = C(k)$. Differentiating it once and twice at $k = 0$,

$$\partial_k B + \partial_w B\, w' = C', \qquad \partial_{kk} B + 2\,\partial_{kw} B\, w' + \partial_{ww} B\,(w')^2 + \partial_w B\, w'' = C'',$$

and solving for the two unknowns,

$$w'(0) = \frac{C'(0) - \partial_k B}{\partial_w B}, \qquad w''(0) = \frac{C''(0) - \partial_{kk} B - 2\,\partial_{kw} B\; w'(0) - \partial_{ww} B\; w'(0)^2}{\partial_w B}. \tag{35}$$

The five ATM Black partials are classical; with the shorthands $a = \sqrt{w_0}/2$, $\Phi_a = \Phi(a)$, $\varphi_a = \varphi(a)$,

$$\begin{aligned} \partial_k B &= -(1 - \Phi_a), \quad & \partial_w B &= \tfrac{\varphi_a}{2\sqrt{w_0}}, \quad & \partial_{kk} B &= -(1 - \Phi_a) + \tfrac{\varphi_a}{\sqrt{w_0}}, \\ \partial_{kw} B &= \tfrac{\varphi_a}{4\sqrt{w_0}}, & \partial_{ww} B &= -\tfrac{\varphi_a}{4 w_0^{3/2}} - \tfrac{\varphi_a}{16\sqrt{w_0}}. \end{aligned} \tag{36}$$

Finally, translate total variance into volatility: $\sigma(k) = \sqrt{w(k)/\tau}$, so one more application of the same chain rule yields the handles,

$$\sigma_0 = \sigma(0) = \sqrt{\frac{w_0}{\tau}}, \quad s_0 = \sigma'(0) = \frac{w'(0)}{2\sqrt{\tau w_0}}, \quad \kappa_0 = \sigma''(0) = \frac{w''(0)}{2\sqrt{\tau w_0}} - \frac{w'(0)^2}{4\sqrt{\tau}\, w_0^{3/2}}. \tag{37}$$

Every quantity along the route is closed-form or one scalar root, so the handles are exact for every LQD slice: the level in volatility units (annualized on the active clock τ), the skew in vol per unit log-moneyness, the curvature in vol per unit log-moneyness squared.

Remark 2 (One triple, two unit systems)**.** Production carries this information in two *equivalent* coordinate systems: the graph-facing carrier $(\sigma_0, s_0, \kappa_0)$ that Note 14 propagates, and the internal chart's $H = (w_0, s_0, \kappa_0)$ with the ATM total variance first (`models/lqd/ortho.py`). Since $w_0 = \sigma_0^2 \tau$, the two are a smooth bijection — no information is at stake, only the *units* of the first coordinate. The reason to be pedantic is that a volatility handed where a total variance is expected fails silently; so the note fixes the convention once — H always means the total-variance triple — and the graph adapter converts.

Heuristic

Equation (33) is the distribution speaking directly. $-C'(0)$ is the probability of finishing above the forward — the ATM digital — and the skew in (35) is, at heart, the mismatch between that digital and the $\Phi(d_-)$ the Black model would assign it. The curvature adds one more distributional fact, the ATM density f_0. Level, digital, density: three numbers of the law, three handles of the smile.

8.2 The ATM-orthogonal chart

The handles earn their keep when they stop being diagnostics and become *coordinates*: axes of parameter space that a person can hold. What we want is a chart around the current fit whose first three axes *are* level, skew and curvature, and whose remaining axes deliberately do nothing to them. Building it amounts to answering two questions, and each answer is a subspace.

Fix the reference θ^* (in practice, the freshly fitted slice), write $H(\theta) = (w_0, s_0, \kappa_0) \in \mathbb{R}^3$ for the handle map, and let $J = \partial H/\partial\theta \in \mathbb{R}^{3\times d}$ be its Jacobian at θ^* — computed by central differences, which on this deterministic, smooth quadrature is accurate to $\sim 10^{-9}$.

Question 1: what is the cheapest way to move the handles? A perturbation $\delta\theta$ moves them by $J\,\delta\theta$ to first order, so a prescribed handle move δH means three equations in d unknowns — underdetermined, many solutions. Take the shortest one:

$$\delta\theta = U\,\delta H, \qquad U = J^{\mathsf{T}}(JJ^{\mathsf{T}})^{-1}, \tag{38}$$

the least-norm right inverse of J, so that $JU = I_3$. The three columns of U are the *primary directions*: moving along them changes the handles one-for-one, in handle units, with no wasted motion in parameter space. "Cheapest", though, is a claim relative to a metric, and the default metric is the raw Euclidean one on coefficients — convenient, but *arbitrary*: it is not invariant to rescaling a coordinate, and a unit of a_9 is not economically a unit of $\log A_L$. The defensible metric prices a move by its effect, and the chart now takes one: passing the Gauss–Newton (Fisher) metric $G = J_r^{\mathsf{T}} J_r$ of the calibration residuals (which the solver already reports at convergence) makes $U = G^{-1}J^{\mathsf{T}}(JG^{-1}J^{\mathsf{T}})^{-1}$, the move that hits the requested handles while disturbing the *fitted quotes* least — test-locked to be a right inverse with strictly smaller fit impact than the Euclidean columns. Either way the handle Gram's SVD condition number is computed and carried on the chart, because near-degenerate handle Jacobians occur on thin boards and a trader should see the warning, not the crash.

Question 2: which moves leave the handles alone? To first order, exactly the kernel of J — a $(d-3)$-dimensional space. Collect an orthonormal basis of it as the columns of V; these are the *shape directions*. One warning attaches to their *names*: an orthonormal kernel basis is unique only up to rotation and sign, so "shape direction 1" is a session-local label that can rotate between calibrations — a sound interface for interactive sculpting at a frozen reference, not a persistent trader coordinate. The stable vocabulary for persistent controls is the market's own, and the kernel now speaks it: for a chosen package set (25- and 10-delta risk reversals and butterflies, the var swap), the implementation solves, inside $\ker J$, for the least-norm combination moving *one* package by one unit — handles fixed to first order by construction — and reports the full cross-talk matrix of each control into the others, together with the kernel-restricted package Jacobian's rank and condition (on a thin board the packages simply are not independently movable, and the print says so rather than pretending).

Gluing the two answers together gives the chart:

$$\theta = \theta^* + U\alpha + V\xi, \tag{39}$$

with three coordinates α that read directly as handle moves — $H \approx H(\theta^*) + \alpha$ — and $d-3$ coordinates ξ that reshape wings, shoulders and event convexity while the handles stand still.

First order is not forever, so production closes the gap. To hit a target H^{target} *exactly*, it solves the three-dimensional system $H(\theta^* + U\alpha + V\xi) = H^{\text{target}}$ for α by Newton — and here the construction pays for itself twice, because $JU = I_3$ makes the system's Jacobian the identity at the reference: the natural first guess $\alpha = H^{\text{target}} - H(\theta^*)$ is already excellent, and the preconditioner rarely needs refreshing. This *retarget* is what lets a trader drag ATM level, skew or curvature while the optimizer keeps the shape modes for everything else; fig. 8 shows one move of each kind on the running example.

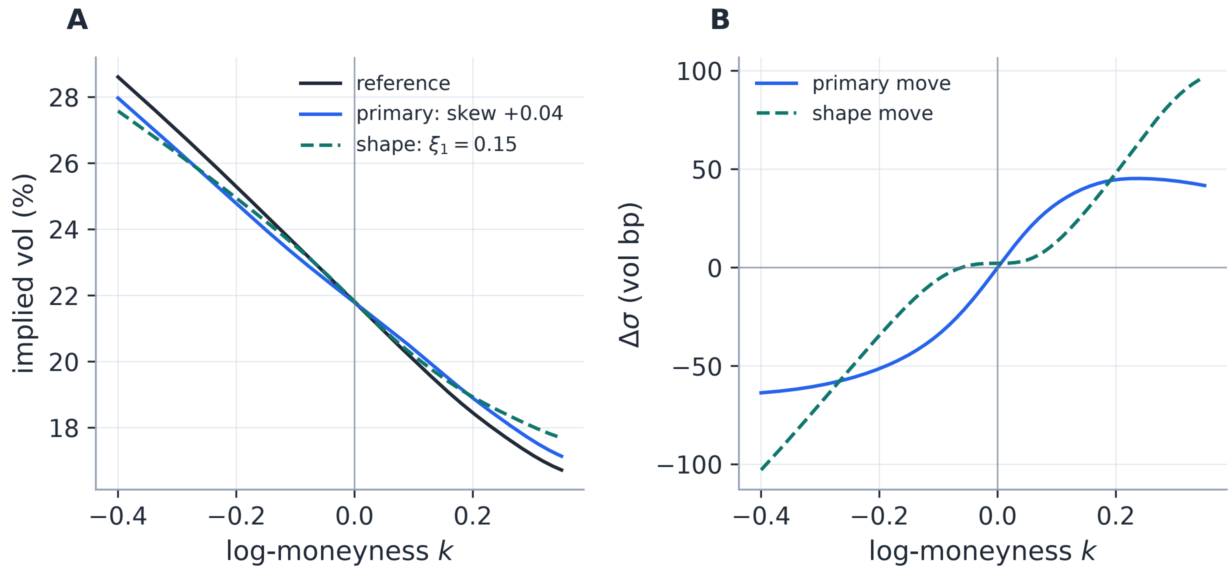

Figure 8: The chart in action on the running SPX-like fit ($d = 10$, so 7 shape directions). Panel A: the reference smile; a *primary* move retargeting the skew by +0.04 exactly (achieved to 6.00×10^{-14} by the Newton solve) with level and curvature held; and a *shape* move along the first kernel direction. Panel B: the same two moves as differences. The primary move tilts the smile through the money; the shape move relocates over 103 vol bp of wing while drifting the ATM level by only 2.17 vol bp, the skew by 7.88×10^{-5} and the curvature by 6.27×10^{-3} — first-order invariance visible to the eye as a flat spot at the money.

Caution

The chart is *local*: U and V are frozen at θ^*, so the handle-invariance of shape directions degrades quadratically with the excursion (the residual drift in fig. 8 is exactly that second order), and the retarget Newton solve can fail — or converge to an *infeasible* point — when the requested handles push the slice toward the $A_R = 1$ wall (a large curvature demand on a thin-tailed reference is the classic case). Callers must treat retargeting as an operation that can fail near the wall, not as a total map on handle space.

9 Calibrating one expiry

We have spent eight sections building a space in which every point is an arbitrage-free slice. Choosing the *right* point is now almost an anticlimax — which was the design goal all along. This section states the objective, the one place it must brake (the wall), where fits start from, and how the shipped objective is actually assembled.

9.1 The core objective

Given quotes (k_i, σ_i) at one expiry, the calibrator solves the bound-free nonlinear least-squares problem (scipy `trf`)

$$\min_{\theta} \sum_i r_i(\theta)^2 \; + \; \lambda \sum_{n \geq 4} n^{2r} a_n^2, \tag{40}$$

with r_i the vega-normalized residuals (31). (An optional per-quote weighting — Note 07's `weightScheme` — multiplies the residuals; it changes nothing below and we suppress it.) The high-order ridge $\lambda\, n^{2r} a_n^2$ is optional; it suppresses oscillation in sparse markets while deliberately leaving the first body modes a_2, a_3 — the workhorses of curvature and asymmetry — untouched. Note what is *absent*: there is no density-positivity term anywhere, because positivity is structural (proposition 2). The one admissibility condition is the wall (14) — and the *chart* the solver runs in decides what it costs. The default solve works in the logistic coordinates of section 5.1, where every trial point is admissible and the wall is unreachable; the raw (L, R, a) and endpoint charts remain available, and because the objective is identical in all three, the optimum is chart-independent to solver tolerance (section 5.1 quotes the measured agreement). In the raw charts the wall is enforced as $A_R < 1 - \varepsilon_R$ with a tiny numerical buffer; in every chart the smooth barrier below remains part of the objective.

The same objective also settles a question about the coefficients themselves that belongs here and not in a footnote: the fitted θ is *not* always unique even when the fitted smile is. Away from the quotes the likelihood is flat — Note 13's subject — and at high order the ridge, not the data, pins the tail of the coefficient vector. The stability study of section 12.1 runs the production fit from ten randomized starts: every start converges, the objective agrees to a cost spread below 10^{-14}, and the *liquid* observables coincide — while the tail parameters spread exactly as far as their weak identification allows. Coefficient non-uniqueness is thus a statement about which *functionals* of the fit are determined, and the honest unit of account for a parametric fitter is the functional, not the coefficient.

9.2 The soft turn before the hard wall

To keep the residual smooth as the optimizer approaches $A_R = 1$, the stacked residual carries a softplus barrier row

$$b(\theta) = \log\bigl(1 + e^{\gamma\,(A_R - c)}\bigr), \tag{41}$$

with centre $c = 0.90$ and steepness $\gamma = 50.0$ (both `FitSettings` coefficients; evaluated as logaddexp$(0, \cdot)$, so a wild trial θ cannot overflow it). The barrier is negligible for $A_R \ll c$ and rises steeply near the wall, turning the fit around *before* the hard rejection; if a trial θ does cross $A_R \geq 1$ (possible only in the raw charts), the price block is replaced by a large finite penalty increasing smoothly in A_R, so the trust region still sees a gradient pointing back to feasibility. In the logistic chart the barrier's role shifts from feasibility device to *economic prior*: no trial can cross the wall, but a fit drifting toward $A_R \approx 1$ is quoting a near-infinite forward-moment budget, and the barrier prices that implausibility — same rows, better job title. The rejection that remains in every chart is the interior overflow budget of section 7.2, which no coordinate change can remove because it guards the arithmetic, not the model.

9.3 Initialization and warm starts

The only cold-start initializer in production is the exactly solvable slice of section 6: $a_n = 0$, $L = R = \log s$ with $s = \sqrt{3w_0}/\pi$ from the variance match, seeded from the ATM-interpolated quote variance. Sequential workflows do better than cold: the background Calibrate sweep *warm-starts* each expiry from the previous, shorter-dated expiry's freshly fitted parameters (adjacent maturities have nearly the same smile, so `trf` lands close to the optimum), and the service can also run a *data-only prepass* — a persistence-free fit used as the measurement stage of the prior and filter machinery (Notes 13 and 15). Two honest caveats on warm starts. Raw coefficients carry the *level* of variance, so a warm start across a large maturity gap inherits the wrong diffusive scale; the endpoint chart softens this (its tail coordinates move smoothly in maturity) but a principled $\sqrt{\tau}$-rescaling of the seed is a roadmap item, as are wing-aware and quantile-projection initializers. And a warm start is a prior in disguise: around an event, yesterday's unimodal seed can shepherd the fit into the wrong basin, which is why the multi-start protocol of section 12.1 exists as a harness — on the running example and the live validation node it found a single basin (ten starts, zero failures), and that finding is a measurement to be repeated on stressed boards, not a theorem.

9.4 The full residual stack

The two-term objective (40) is the lecture version. The shipped objective is a *stack* of residual blocks, and table 3 is the map of the whole territory: the data block and ridge we have met, the calendar slack of section 11, the barrier, and then a run of optional blocks — var-swap targets, priors, filter terms — each of which belongs to its own note in the series and simply plugs extra rows into the same least-squares problem. Two properties of the assembly are worth carrying away. Every optional block is byte-identical-to-absent when unused (test-locked), so switching a feature off really is off. And the Jacobian column matters operationally: the analytic route of section 10 covers the first five blocks *and* both var-swap rows (they ride the same sensitivity pass), while activating a prior-anchor or operator block switches the whole fit to scipy's finite-difference Jacobian — correct, merely not accelerated.

Residual block	Active when	Units / scaling	Jacobian
Mid data block	`fitMode=mid` (default)	vega-normalized price ($\approx$ vol)	analytic
Band data block (hinge + mid anchor; Note 07)	`fitMode` `bidask/haircut`	vega-normalized price	analytic
High-order ridge	`regLambda`> 0 (rows $n \geq 4$)	coefficient units	analytic
Calendar slack (Note 10)	`enforceCalendar` + a nearer fitted expiry	upper share (normalized price)	analytic
A_R soft barrier	always (one row)	dimensionless	analytic
Market var-swap (Note 08)	`varSwapEnabled` + an active quote	vol	analytic
Prior strike-gap anchor (Note 13)	`strike_gap` / `hybrid` tail, active prior	vega-normalized price	FD fallback
Prior var-swap companion (Note 13)	alongside an active prior	vol	analytic
Operator prior (Note 13)	`quote_operator` / `hybrid`, gated per operator	operator units (vol)	FD fallback
Active-filter MAP (Note 15)	filter mode `active` (rides the operator block, ungated)	handle units	FD fallback

Table 3: The LQD residual stack as production assembles it (`models/lqd/calibrate.py`; prior/filter targets are resolved in `api/service.py`). "FD fallback" means the presence of that block switches the entire fit to the finite-difference Jacobian — since this revision only the prior-anchor and operator blocks do; both var-swap rows ride the analytic pass (section 10).

9.5 The native var-swap route

The log-contract identity gives the var-swap proxy in one line of the model's own coordinates:

$$w_{\mathrm{vs}} = -2\,\mathbb{E}[X] = -2\int_{-\infty}^{\infty} Q(z)\,u(z)\big(1-u(z)\big)\,\mathrm{d}z, \tag{42}$$

a single quadrature on the grid the slice already owns, with integrand decaying like $z\,e^{-|z|}$ (`quadrature.py::var_swap_strike`). Two qualifications keep the claim honest. First, this native route is dramatically cheaper than the generic strike replication of Note 08 — which would re-solve implied variance on a strike grid at every Jacobian column — but activating the var-swap block also forfeits the analytic Jacobian (table 3), so a var-swap fit is not literally as cheap as a mid fit. Second, the identity prices the *log contract*: its correspondence with a realized-variance swap assumes continuous monitoring of a continuous (diffusive) path; jumps and discrete sampling take the usual corrections, outside this model's scope.

10 The derivative that comes free

The dominant cost of (40) used to be the Jacobian: scipy rebuilt the entire quadrature pipeline $P+1$ times per iteration to finite-difference it, with $P = N+1$ the parameter count. The Vol-Fitter instead propagates the analytic Jacobian of the core residual stack — data block (mid or band), ridge, calendar slack, barrier — in *one* quadrature pass. The key is a cancellation identity, and it is the cleanest piece of mathematics in the implementation.

10.1 The cancellation identity

The subtlety is that the priced call (28) depends on θ twice: explicitly through the curves Q and G, and implicitly through the strike root $z_k(\theta)$ solving $Q(z_k;\theta) = k$. A naive gradient must differentiate the root. Differentiate the pricing identity and watch:

$$\frac{\mathrm{d}C}{\mathrm{d}\theta} = \underbrace{\frac{\partial G}{\partial \theta}\Big|_{z=z_k}}_{\text{explicit}} + \Big[\,G'(z_k) + e^{k}\,u'(z_k)\Big]\frac{\mathrm{d}z_k}{\mathrm{d}\theta}. \tag{43}$$

Now $G'(z) = -e^{Q(z)}u(1-u)$ from (27), while at the root $e^{Q(z_k)} = e^k$ and $u'(z_k) = u_k(1-u_k)$. The bracket is

$$-e^k u_k(1-u_k) + e^k u_k(1-u_k) = 0. \tag{44}$$

Theorem 1 (Implicit-root cancellation)**.** *The implicit dependence of the strike root on the parameters drops out of the price gradient:*

$$\frac{\mathrm{d}C}{\mathrm{d}\theta} = \frac{\partial G}{\partial \theta}\Big|_{z=z_k}. \tag{45}$$

Remark 3. This is an envelope argument in disguise. $C(k)$ is an integral whose lower limit z_k sits exactly where the integrand of the combined asset-and-cash ledger vanishes — the payoff $(e^Q - e^k)$ is zero at its own exercise boundary — so moving the boundary is free, to first order. The same mechanism makes the American exercise boundary drop out of American option Greeks.

So $\mathrm{d}C/\mathrm{d}\theta$ is obtained by evaluating the *nodal* parameter sensitivities of the upper share at the fixed point z_k — by the very Hermite interpolation used to price, fed $\partial G/\partial\theta_j$ and $\partial G'/\partial\theta_j$ instead of the nodal values.

10.2 Nodal sensitivities in one pass

With the root out of the way, there is nothing left to invent: we simply differentiate the five-step recipe of section 7.2, step by step, and every step stays a cumulative integral. Two structural facts make it painless. First, g is *affine* in θ, with constant basis $\phi_j = \partial g/\partial\theta_j$ — namely $\phi_L = 1-u$, $\phi_R = u$, $\phi_{a_n} = P_n(1-2u)$ — so differentiating anything of the form e^g just multiplies it by a fixed, cached array. Second, integration and $\partial/\partial\theta_j$ commute, so a derivative of a cumulative integral is the cumulative integral of the derivative.

Walk the recipe. Step 2 (the unshifted quantile $\overline{Q} = Q - \mu$) differentiates under the integral sign,

$$\frac{\partial\overline{Q}(z)}{\partial\theta_j} = \int_0^z e^g\,\phi_j\,\mathrm{d}s, \tag{46}$$

one more cumulative Simpson pass per parameter. Step 3 (the shift $\mu = -\log M$, with M the martingale integral) is a logarithm, so

$$\frac{\partial\mu}{\partial\theta_j} = -\frac{1}{M}\int e^{\overline{Q}}\,\frac{\partial\overline{Q}}{\partial\theta_j}\,u(1-u)\,\mathrm{d}z \;+\; (\text{tail-correction terms}), \tag{47}$$

where the corrections (29) contribute through $\overline{Q}(\pm Z)$ and through A_L, A_R — whose own gradients cost nothing, since (7) gives $\partial A_L/\partial\theta = A_L\,(1,0,1,1,\dots)$ and $\partial A_R/\partial\theta = A_R\,(0,1,1,-1,\dots)$ with signs $(-1)^n$. Step 4 (the share) is another cumulative pass: with $\partial Q = \partial\mu + \partial\overline{Q}$ in hand, integrate $\partial(e^Q u(1-u))/\partial\theta_j$ from the right, add its right-tail correction, and note that the nodal slope sensitivity $\partial G'/\partial\theta_j$ is the same integrand negated — exactly the pair the Hermite interpolant wants. The non-price rows of the stack then differentiate in one line each: the band hinge multiplies $\mathrm{d}C/\mathrm{d}\theta$ by the violation sign, the ridge is diagonal on the a-columns, the calendar slack reuses the share sensitivity at the constraint nodes, and the barrier row is $\operatorname{expit}\big(\gamma(A_R - c)\big)\,\gamma\,\partial A_R/\partial\theta$. One quadrature pass, all P columns.

One honesty note on exactness. Theorem 1 is exact for the *continuous* objects. The discretization interpolates Q and G as two separate Hermite interpolants, so between nodes the interpolated price is not algebraically the integral of the interpolated quantile, and the implemented gradient inherits that (tiny) mismatch. The operational claim is therefore *benchmarked numerical agreement*, not algebraic identity — and it is benchmarked twice over, in fig. 9.

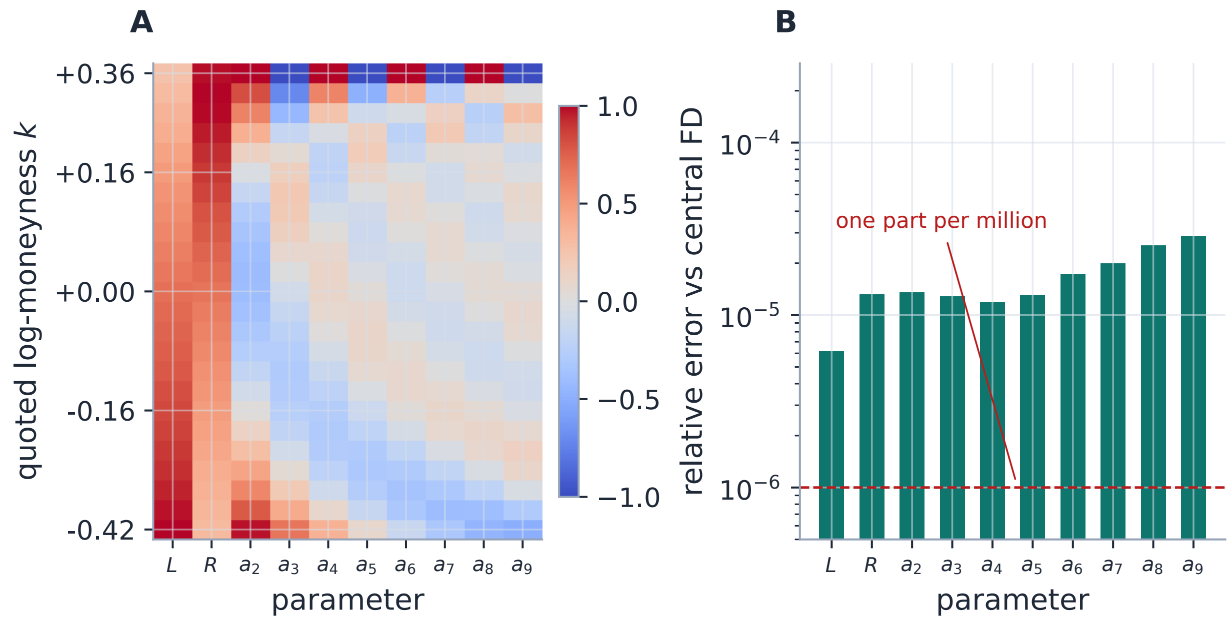

Figure 9: The analytic Jacobian audited against an independent central finite difference on the running fit (10 parameters). Panel A: each column is one parameter's price sensitivity across the strike ladder, normalized per column to $[-1, 1]$ (the colour scale) — L and R weight the wings most heavily, each Legendre mode paints its own oscillation, and (read the top rows of the body columns) every a_n also reaches the wings through the endpoint coupling of (7) — the raw chart's signature, decoupled in section 5.1. Panel B: column-wise relative disagreement between the one-pass analytic Jacobian and the central difference: at most 2.88×10^{-5}, i.e. agreement to about five significant digits with a derivative computed by a completely different route.

10.3 Scope and speed

The analytic route covers every configuration without prior-anchor or operator blocks (table 3); those two fall back to scipy's two-point finite difference, correct but not accelerated. The var-swap rows were the cheapest piece and are now in: $\partial w_{\mathrm{vs}}/\partial\theta = -2\int(\partial Q/\partial\theta)\, u(1-u)\,\mathrm{d}z$ needs only the nodal quantile sensitivity the pass already computes, one extra chain factor $\mathrm{d}\sigma_{\mathrm{vs}}/\mathrm{d}w = 1/(2\sqrt{w\tau})$, and nothing else — so the common var-swap-enabled configuration keeps the analytic path (FD-audited like every other block). Measured on this repository's development machine, the analytic Jacobian reaches the *same* optimum as the finite difference (costs agreeing to $\sim 10^{-11}$ relative, parameters to $\lesssim 10^{-5}$) at 1.44–1.97× the speed across $N = 6 \ldots 12$ — but wall-clock ratios are machine-, cache- and configuration-dependent, so the measurement protocol, the timing table and the chart — generated *from the same single run* — live in appendix B. The structural point is machine-independent, and worth stating precisely: the analytic pass is $O(P\, n_{\mathrm{grid}})$ — one cumulative sensitivity integration per parameter riding one build — versus the finite difference's $P{+}1$ *complete* builds with their normalizations and root solves. It removes the rebuilds, not the linear-in-P work.

11 Many expiries: order the ledgers

One dimension of the problem has been silent so far: time. Proposition 2 silences butterflies *within* a slice, but a surface is many slices, and fitting them independently leaves a second static arbitrage on the table: nothing yet prevents a longer-dated call from pricing *below* a shorter-dated one at the same log-moneyness — a calendar spread the market would pay you to own. (Two standing assumptions frame everything here: carry is deterministic, so normalized units under each expiry's forward measure are legitimate simultaneously; and every slice is normalized to mean one.) For expiries $\tau_i < \tau_j$, absence of calendar arbitrage is the requirement $C_i(k) \le C_j(k)$ at every k — in the language of probability, the laws of Y must increase in *convex order* [7]: the longer-dated law is more spread out, worth at least as much against every convex payoff, without its mean moving. Let the section's conclusion be stated before its machinery, because it scopes everything that follows: what production enforces is *control, not a theorem* — finite-grid, finite-weight, optional — and the paragraph closing this section says exactly what that means and what is reported when control falls short.

Checking a continuum of call inequalities sounds like work. The pleasant surprise is that the model already owns the right test object, because the call curve and the upper share are *Legendre transforms of each other*. For any rank u and any log-strike k,

$$G(u) - e^k(1-u) = \int_u^1 \left(e^{Q(v)} - e^k\right) \mathrm{d}v \;\le\; C(k), \tag{48}$$

because extending the integral to where the integrand is negative can only lose money — with equality exactly at $u = u_k$, the strike's own rank. (That equality *is* the pricing identity (28), rediscovered as a maximization.) Read once in each direction:

$$C(k) = \max_{u\in[0,1]} \left[G(u) - e^k(1-u)\right], \qquad G(u) = \min_k \left[C(k) + e^k(1-u)\right]. \tag{49}$$

Each curve is an extremum over the other, and pointwise domination passes through maxima and minima in both directions. Hence, for mean-one slices,

$$C_i(k) \le C_j(k) \;\text{ for all } k \qquad \Longleftrightarrow \qquad G_i(u) \le G_j(u) \;\text{ for all } u \in [0,1], \tag{50}$$

with equality at $u = 0$, where both shares equal the common mean, one. A continuum of option inequalities has become the pointwise ordering of two curves that are already on the grid — the same ledger that priced the calls and carried the Jacobian, now doing its third job.

Enforcement follows the geometry. The surface is built nearest-to-farthest: each new slice is fitted subject to $G_j(u_m) \ge G_i(u_m)$ on a dense rank grid (per-node slack tolerance zero in production), as the soft calendar block of table 3 with penalty weight `calendarWeight`; the constraint is evaluated at z-values, so it is exact whatever grid the optimizer runs on (section 7.2). Rank space is also the better-conditioned place to test: no implied-volatility inversion at any strike, and full meaning retained in the far wings, exactly where inversion goes blind.

Figure 10 shows the whole story on fitted slices, and it is worth reading deliberately, panel by panel. Panel A draws the ledgers themselves for a legitimate pair: the running example as the far expiry, and a quarter-year slice fitted to the same smile shape with proportionally smaller variance as the near one. Both curves start at $G(0) = 1$ — the whole mean, since every outcome lies above the 0th percentile — and both fall to zero at $u = 1$. In between, the far curve rides everywhere *above* the near one, and the shaded sliver between them is the calendar's entire economic content: at every rank, the top $(1-u)$-tranche of the longer-dated law carries more of the mean, which by (49) is precisely the statement that every call is worth more at the longer expiry. Note how thin the sliver is — a legitimate term structure clears the bar by little, which is why calendar violations are so easy to manufacture and so hard to spot by eye.

Panel B plots that sliver directly: the calendar gap $G_{\text{far}}(u) - G_{\text{near}}(u)$, which the order (50) requires to be non-negative for every rank. The gap must vanish at both ends (at $u = 0$ both ledgers equal the common mean; at $u = 1$ both are empty), so all the information lives in the middle, and three fits are drawn there. Green, the legitimate pair: an arch that stays non-negative everywhere (minimum 0.00087). Red, a deliberately crossed pair — the far slice refitted to quotes whose body variance sits 10% *below* the near slice while its wings sit above: the gap dips to -0.00218 around the $u \approx 0.39$ rank (shaded), and that dip is a concrete trade, a calendar spread at the corresponding strikes that the crossed surface prices *negative*. Each smile on its own looks perfectly respectable; only the ledger comparison exposes the crossing. Dashed blue, the same crossed quotes refitted *with* the production soft floor active: the fit concedes what the quotes demand, the gap is pressed against zero from below, and the residual violation is -3.45×10^{-6} — four orders of magnitude smaller, but not exactly zero, because the floor is a finite-weight hinge, not a hard constraint. That residual is the honest signature of *soft* enforcement, and it is the subject of the closing paragraph.

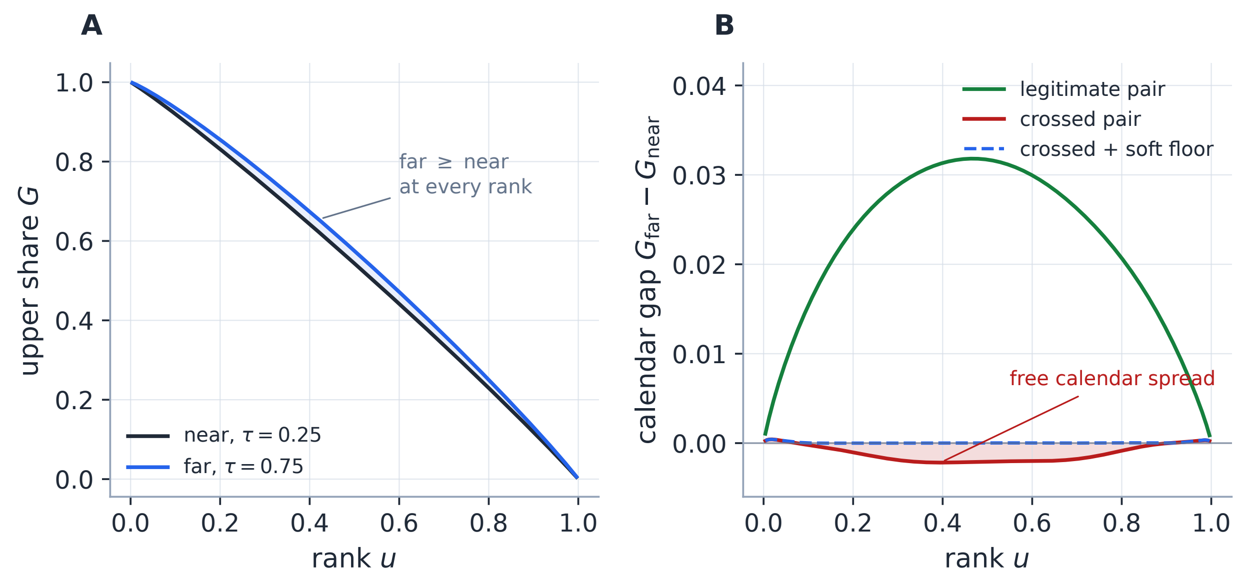

Figure 10: Convex order read off the ledger. Panel A: the upper shares of a legitimate near/far pair; the shaded sliver between the curves — far above near at every rank — is exactly the value ordering of every calendar spread. Panel B: the calendar gap for three fits: the legitimate pair (green, minimum 0.00087), a crafted crossed pair (red; the shaded dip to -0.00218 is a free calendar spread hiding between two respectable-looking smiles), and the crossed quotes refitted under the production soft floor (dashed blue, residual violation -3.45×10^{-6}: pressed to zero, softly).

Honesty about the guarantee, as promised at the top: this is *control*, not a theorem, in four specific ways. The constraint is checked on a finite grid, so violations can hide between nodes (bounded, in practice, by the same interpolation tolerance section 7.4 audits). The hinge carries a finite weight, so a hard-pressed fit can retain a small violation rather than destroy the quote fit (the dashed curve of fig. 10 retains exactly -3.45×10^{-6}). A sequential sweep constrains each expiry against an already-fitted neighbour, so refitting an early expiry can stale every later constraint — which is why the production surface path screens the finished ladder and *jointly* repairs any violation-connected group of slices, rather than trusting the one-way pass. And the whole coupling is a user toggle, so off means off. Residual violations are measured and reported by the surface diagnostics rather than assumed away — and, as of this revision, in the units a desk prices: alongside the normalized-price violation, each node reports the worst-violation strike K^* (the cheapest sell-near/buy-far calendar spread), the violation in currency per share, in option-price ticks (when the source chain quotes a tick), and as a fraction of the local bid–ask price width at K^* — so “violation 10^{-6}” resolves into “a tenth of a tick, a fortieth of the spread: unharvestable” or into a number that stops a publish. Note 10 develops the model-agnostic calendar machinery (and the subtlety that the variance floor must be confined to the traded strike range); here we record only that for LQD the natural constraint object is the ledger the model already owns.

12 Case files

A lecture should end with the machine run end-to-end, and with the operator saying plainly what the run does and does not prove. Both cases are fitted by the production calibrator — not a notebook re-implementation — against two deterministic targets; of the note’s generators, `gen_lqd_fresh.py` computes the quoted diagnostics, `gen_lqd_geometry.py` draws the figures from the identical fits, and `gen_lqd_referee.py` runs the stability protocol below. What they prove is *approximation capacity*: that the basis can reproduce a known clean curve through option prices and back. What they do not prove is market fit; that evidence, against noisy NBBO quotes, lives in the backtest reports (section 1). Table 4 puts the two side by side.

Table 4: The two case files at a glance (fresh production run).

Diagnostic	SPX-like	Event
Model order	9	16
Quotes	24	37
Max error (vol bp)	1.341	2.444
RMS error (vol bp)	0.408	0.999
A_L	0.157473	0.015531
A_R	0.037786	0.014044

12.1 Case file: the SPX-like strip

The running example in full. The target is a skewed SSVI-shaped curve at $\tau = 0.75$ (appendix C) with an ATM volatility near 21.7961% and a pronounced index put skew, sampled at 24 irregular strikes and perturbed by a deterministic sub-two-bp ripple — so the fit is a genuine fit, not a sterile recovery of a noiseless curve. A 9-mode LQD slice converges in 8 residual evaluations with

$$\text{max error} = 1.341 \text{ vol bp}, \qquad \text{RMS} = 0.408 \text{ vol bp}, \qquad \mathbb{E}[e^X] - 1 = -1.11 \times 10^{-16},$$

exact ATM handles $\sigma_0 = 21.7961\%$, $s_0 = -0.177846$, $\kappa_0 = 0.002673$, endpoint scales $A_L = 0.157473$, $A_R = 0.037786$, Lee slopes $\beta_L = 0.073087$, $\beta_R = 0.019259$, and a native var-swap strike (42) of 22.9685% — above the ATM volatility, as the skew demands. Figure 11 shows the fit, its error ledger, and what the skew looks like as probability; fig. 12 follows the fitted endpoint scales through the maps of section 4.

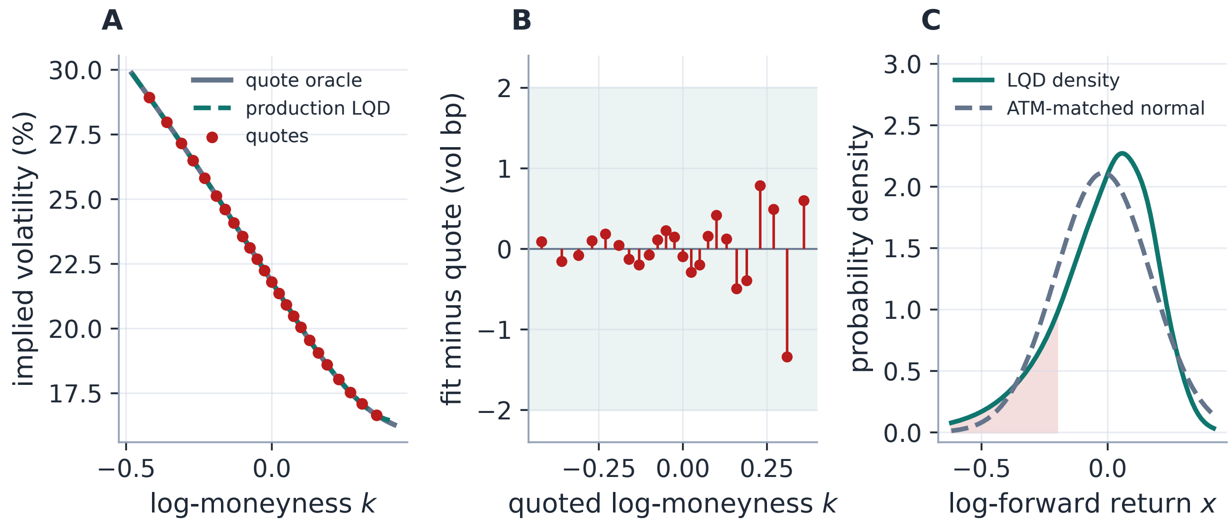

Figure 11: The SPX-like case. Panel A: quote oracle, quotes, and the production fit. Panel B: the error ledger at the quotes — every residual inside ± 2 vol bp (shaded), maximum 1.341 vol bp against quotes carrying a deliberate ~ 2 bp ripple. Panel C: the fitted density against an ATM-matched normal: the entire content of the skew is the transfer of mass into the left tail (shaded), bought by a thinner right shoulder. Beyond the quoted window the curve is the model's own Lee-consistent extrapolation, not an oracle continuation.

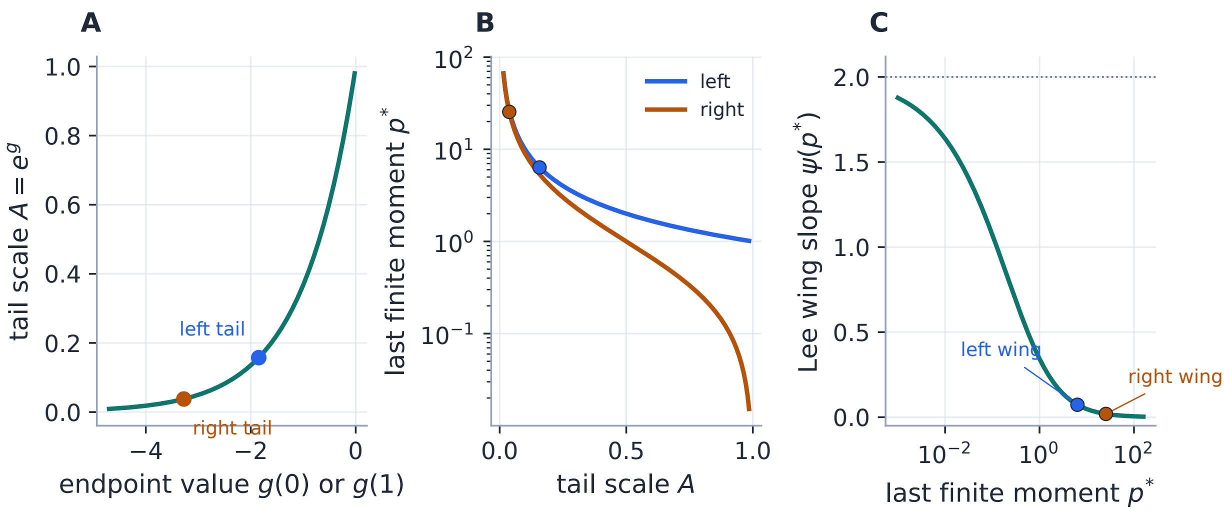

Figure 12: The fitted tails walked through the three links of section 4: endpoint value of g to tail scale ($A_L = 0.157473$, $A_R = 0.037786$), tail scale to last finite moment (13), and moment to Lee wing slope (16) ($\beta_L = 0.073087$, $\beta_R = 0.019259$). No explicit wing prior was imposed — but "implied by the quotes" would claim too much: these numbers are what the quotes, the basis order, the ridge, the vega weighting and the exponential skeleton *jointly* selected, and the stability study below measures how much of each number is data and how much is prior.

The tail-stability study. The committee question every tail number must survive is: *how much can it move while every liquid quote stays respectable?* The protocol is the referee's own: jackknife the outer one and two quotes per side, sweep the order N over 6–16 and the ridge λ over 0–10^{-2}, perturb all quotes by i.i.d. 1 vol bp noise twenty times, and refit from ten randomized starts — several dozen production refits, each fully converged. Figure 13 draws every observable's excursion; the asymmetry between its rows is the finding.

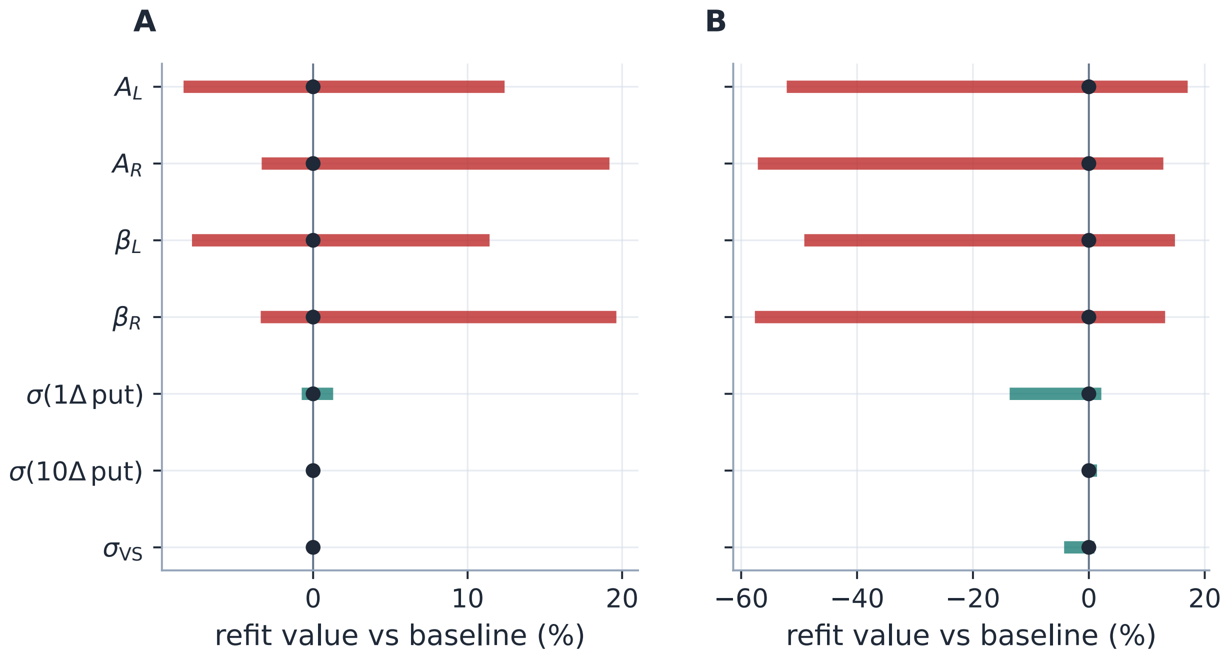

Figure 13: What the quotes pin, and what the prior chooses. Each row is one observable; the bar is its full min–max excursion across the whole perturbation protocol, relative to the baseline fit (dot at zero); red rows are asymptotic tail parameters, teal rows are liquid observables. Panel A, the running example: the var-swap moves 0.2%, the 10Δ put vol 0.3% — while A_R roams 22.5% and the Lee slope β_R 23.0%. Panel B, a live SPY node from a stored market validation (September expiry, 240 quotes, bid–ask band fit): the same shape, louder — tails spread 70.0% while the 10Δ put vol moves 1.5%. The tail ladder is a *prior-dominated* output and must be quoted with these fans, not as point estimates.

The same study answers the multi-start question cleanly (ten starts, zero failures, cost spread below 10^{-14} on both the synthetic and the live strip: one basin), and it calibrates how to *read* the wings. Traders do not trade the $k \to \infty$ limit; they trade 10Δ and 1Δ strikes, and fig. 14 shows the effective variance slope $w(k)/|k|$ still 2.68× the Lee limit at the 10Δ put and 1.72× at the 1Δ put. The asymptotic slope is a boundary condition on extrapolation, not a quote at any tradable strike — which is precisely why its wide fan in fig. 13 coexists with basis-point-stable delta-wing vols.

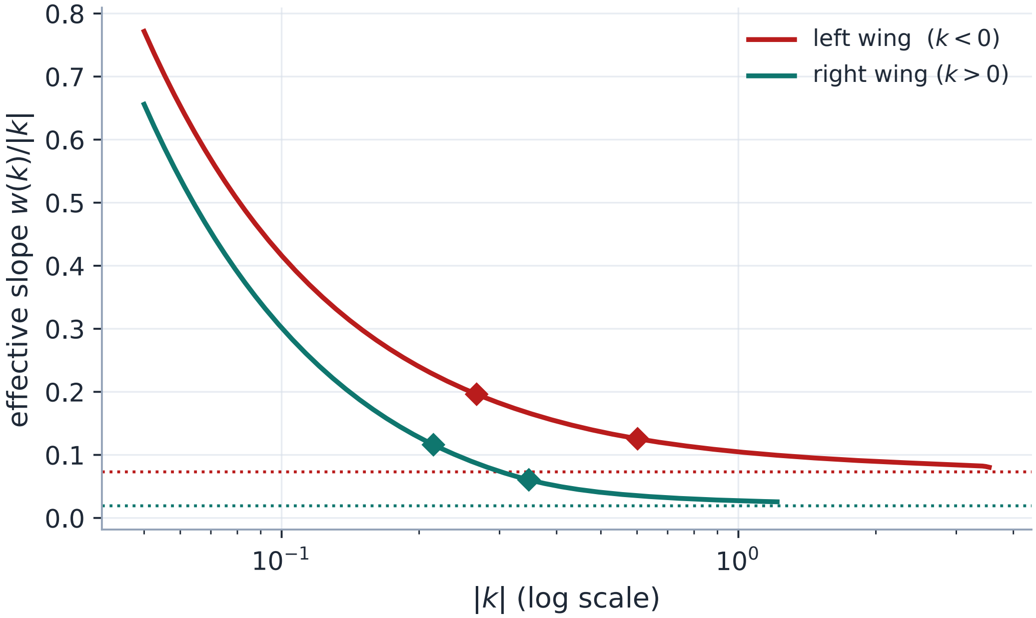

Figure 14: The limit is not the wing. Effective slope $w(k)/|k|$ of the fitted running example against $|k|$ (log scale), both wings, with the Lee limits dotted and the $10\Delta/1\Delta$ strikes marked (diamonds). The curves END where double-precision Black inversion stops resolving the option price — the asymptote has not been reached at any strike a computer, let alone a market, can quote. Effective-to-limit ratios at the marks: 2.68 (10Δ put), 1.72 (1Δ put), 6.03 (10Δ call), 3.14 (1Δ call).

12.2 Case file: the density wears two hats

Ahead of a scheduled binary event — earnings, a court ruling, a referendum — the terminal density can be *bimodal*: the market prices two distinct landing zones. A low-order direct-volatility model fights this shape; a quadratic-in-k smile cannot make a central trough with supported wings. The target here is an asymmetric mixture of two log-return normals (weights 56.0%/44%, centres -0.074804 and $+0.085196$ after martingale normalization, distinct widths; appendix C) at $\tau = 24/365$, quoted at 37 strikes. A 16-mode fit — the top of the schema range, an event slice's natural habitat — recovers both the smile and the two-humped density (fig. 15) with maximum error 2.444 vol bp, endpoint scales $A_L = 0.015531$, $A_R = 0.014044$, and $\mathbb{E}[e^X] - 1 = -2.22 \times 10^{-16}$. A sharp look at panel A corrects a tempting misreading: the *implied-vol curve* of this bimodal law is a single central hump, not a W — bimodality announces itself in the smile only as subtle shoulder structure, and lives plainly in the density. That is exactly why the title of this case file says the density wears two hats: the two charts carry the same information at very different legibility. Positivity of the density is automatic even where the high Legendre modes carve the trough; what the basis *cannot* do is place an exact atom — the hats are smooth by construction (section 14). In a sparse live market one would lean on the ridge or fix the handles through the chart of section 8.2 to keep the extra modes from fitting noise.

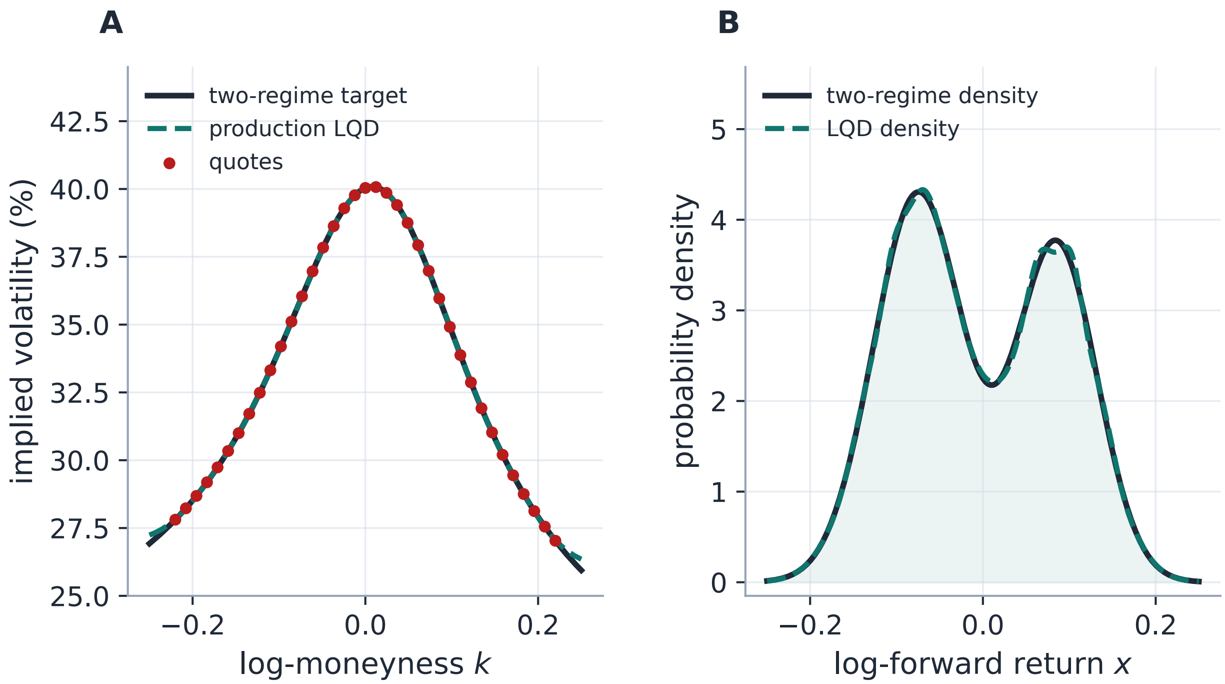

Figure 15: The event case. Panel A: the two-regime target smile and the 16-mode production fit — a single-humped implied-vol curve whose only smile-side trace of bimodality is in the shoulders. Panel B: the true mixture density and the LQD density: both hats, the asymmetry, and the trough survive the round trip through option prices, with positivity never at risk.

Approximation-capacity claims deserve a control group and a latent-object error, not only a quote error, so fig. 16 refits the same strip at $N = 6$. On the *smile* the comparator looks nearly respectable — maximum error 28.3 vol bp against the high-order fit's 2.44 — but on the *density* it overshoots the left hat and half-fills the trough: quote error L^1-density error, and mode placement are three different ledgers (here: L^1 distance 0.059 vs 0.015; worst mode displacement 90 vs 123 bp of return). A fitter judged only at its quotes can be quietly wrong between them — the recurring lesson of this series, and the reason case files here always report the latent object.

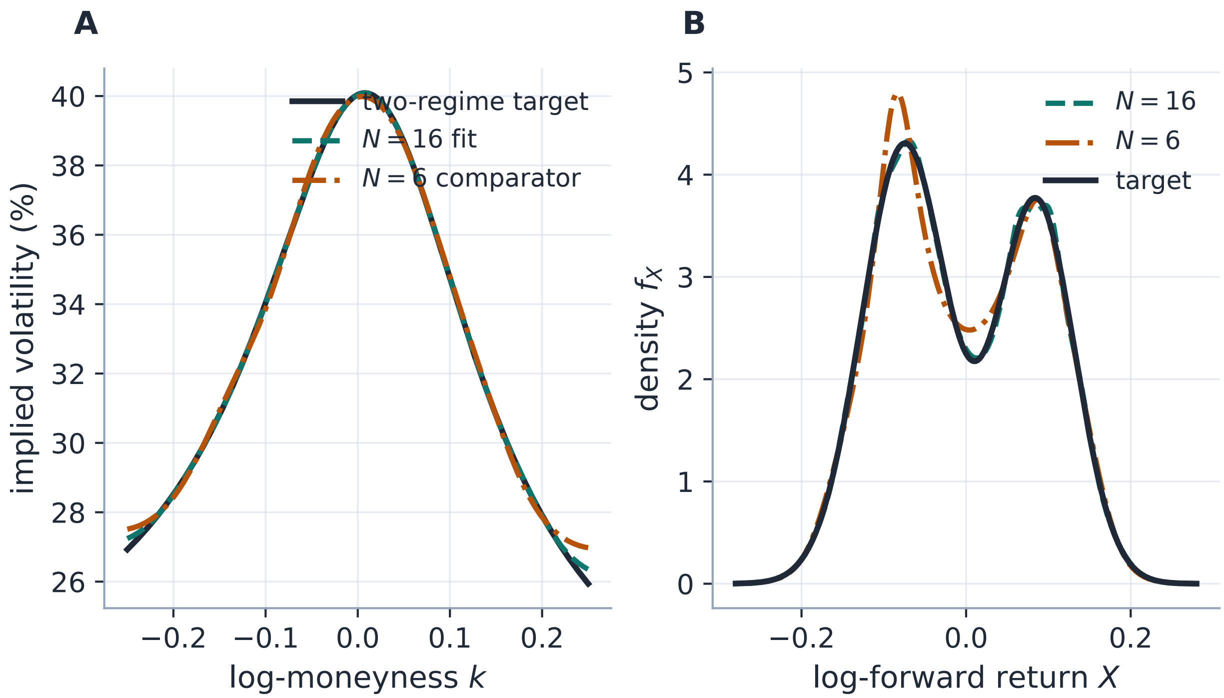

Figure 16: What the extra modes buy. Panel A: the event target with the $N = 16$ production fit and an $N = 6$ comparator — in implied vol the three curves nearly coincide, and nothing here looks alarming. Panel B: the same three objects as densities — the comparator (dash-dot) overshoots the left hat and misses the trough's depth. The information the low order lacks is invisible in the chart traders quote and obvious in the chart risk systems differentiate; capacity must be judged on both.

13 What is adopted, and what is original

Intellectual honesty starts with the ledger of debts. The log-quantile-density transformation itself is established statistics: Petersen and Müller [4] use exactly this transformation to map densities to an unconstrained function space for functional data analysis, and the quantile-parameterized distribution literature — Keelin's metalog family [5] most visibly — builds flexible laws on logistic-quantile skeletons closely related to (5). Lee's moment analysis is classical, and Breeden–Litzenberger older still. What this note contributes is the *option-pricing assembly* of those pieces, and three parts of the assembly are, to our knowledge, worth singling out — with the caveat that a fuller literature comparison has not been made, so "original" here means "not found by us elsewhere", not a priority claim:

1. **The martingale-complete endpoint construction.** The skeleton $-\log u - \log(1-u) + (1-u)L + uR$ with the mean-one normalization solved by a single shift, the endpoint values identified with tail scales, critical moments and Lee slopes (16), and (as of this revision) the endpoint-decoupled logistic chart of section 5.1 making the admissible set exactly $\mathbb{R}^d$.

2. **The exact ATM handles and the orthogonal chart** (section 8). Level, skew and curvature are closed-form functionals of the quantile, so the model carries genuine trader coordinates *and* a chart in which shape modes do not disturb them — the same three handles the graph extrapolator of Note 14 propagates.

3. **The implicit-root cancellation** (theorem 1) and the upper-share ledger serving pricing, sensitivities and the convex order at once. The envelope-style cancellation is elementary calculus applied in the right coordinates; its *use* — one quadrature pass serving all Jacobian columns, audited to 2.88×10^{-5} against central differences — is the engineering contribution.

14 Limitations

A lecture that sells a model without drawing its boundary is an advertisement. Here is where this one's guarantees stop. The headline first: LQD removes butterfly arbitrage by construction but *not* calendar arbitrage across independently fitted expiries — the coupling of section 11 is soft, finite-weight and optional, so a coupled surface is calendar-*controlled*, not calendar-guaranteed. Beyond that:

- **Near the integrability wall.** As $A_R \to 1$ the forward integral degenerates: the tail corrections (29) dominate, conditioning of the fit and of the retarget Newton solve deteriorates, and the barrier is doing real work. Admissible is not the same as comfortable.
- **Atoms of any kind.** The model is a continuous law on $(0, \infty)$: not only a jump-to-default atom at zero but *any* exact mass point (a pinned settlement price, a tender offer) can only be approximated by a sharp continuous feature, never represented.
- **Positive is not plausible.** Structural positivity guarantees a *density*, not a sensible one: an unregularized high-order fit can produce economically implausible — multi-modal, spiky — yet perfectly arbitrage-free shapes. Positivity removes one failure class; the ridge, band weighting and judgment handle the rest.
- **Coefficient non-uniqueness in practice.** Distinct coefficient vectors can price indistinguishably (near-degenerate directions grow with N and with sparse data; section 9 treats this in the main line), so fitted coefficients are not comparable across fits — compare handles, curves and densities, not raw θ.
- **Tail outputs are prior-dominated.** The stability fans of fig. 13 are part of the model's interface: A_L, A_R, critical moments and Lee slopes must be consumed as *diagnostics with uncertainty*, never as approved point-estimate risk outputs, until wing quotes actually pin them.
- **A marginal is not a dynamics.** Everything here is one expiry's risk-neutral *marginal*; even a full convex-ordered surface does not select the dynamics between expiries. Barriers, cliquets, forward-starts and other path-dependent trades need a model *through* the marginals (local vol, Note 04; spot–vol dynamics, Note 12), and nothing in this note prices them.
- **Garbage forward, garbage skew.** The whole construction lives in $k = \log(K/F)$: a forward or dividend error of δ shifts every quote's moneyness by $-\delta$, and to first order the fitter reads it as a spurious vol move of $(\partial\sigma/\partial k)\,\delta$ plus a density translation — easily misdiagnosed as skew. Forward hygiene (Note 06) is an input contract of this model, not a refinement.
- **Interface gaps, remaining.** The Gauss–Newton chart metric, the package controls and the desk-unit calendar reporting are implemented and test-locked (sections 8.2 and 11), but the interactive front-end still drives the Euclidean chart and generic shape sliders — wiring the

packages and the condition warning into the user interface, and choosing when the GN metric becomes the interactive default, are the remaining product decisions. Hedge-instrument risk metrics beyond Gauss–Newton remain unexplored.

- **Unstudied sensitivities.** There is no dedicated study of handle *drift* under rolling recalibration, nor of digital/one-touch sensitivities to the interpolation scheme; both are open measurement items, not established properties. Equal-footing market benchmarks against SVI/SSVI-class fitters — in-spread rates, held-out strikes, next-snapshot repricing, hedge P&L at matched parameter count and CPU — live in the backtest programme, not in this note, and approval-grade claims should wait for them.

15 Traceability

One caveat on the "match note" anchors: the fixture coefficients those tests lock were recorded when the original benchmark was frozen, so they are *legacy locks* — they pin production behaviour against regression, but their literal values can lag the freshly regenerated numbers in this note (which always come from fresh generator runs). Where the two disagree, the fresh run is the current truth and the test is the regression tripwire.

Table 5: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor *Test anchors*

Density positive; call curve monotone and convex; parity holds	proposition 2	`models/lqd/quadrature.py`
		`test_lqd_pricing.py::test_density_positive`
		`::test_call_curve_monotone_and_convex_in_strike`
		`::test_put_call_parity`

Martingale shift correct; $\mathbb{E}[e^X] = 1$	(26)	`models/lqd/quadrature.py`
		`test_lqd_pricing.py::test_svi_fit_martingale_shift_matches_note`
		`::test_svi_fit_martingale_integral_is_one`

Endpoint scales and Lee slopes; stable for extreme scales	(7)–(16)	`models/lqd/basis.py`
		`test_lqd_basis.py::test_svi_fit_endpoint_scales_match_note`
		`::test_svi_fit_lee_slopes_match_note`
		`::test_lee_slopes_stable_for_tiny_nonzero_endpoint_scales`

Legendre basis: recursion and orthogonality	(18)	`models/lqd/basis.py`
		`test_lqd_basis.py::test_legendre_recursion_matches_explicit_polynomials`
		`::test_legendre_orthogonality_on_unit_interval`

Exact ATM handles	(37)	`models/lqd/atm.py`
		`test_lqd_atm.py::test_atm_level_matches_implied_vol`
		`::test_atm_skew_matches_finite_difference`
		`::test_atm_curvature_matches_finite_difference`

Claim	Object	Code anchor *Test anchors*

Orthogonal chart: shape moves preserve handles; retarget exact	section 8.2	`models/lqd/ortho.py`
		`test_lqd_ortho.py::test_shape_move_leaves_handles_nearly_unchanged`
		`::test_retarget_hits_exact_handles`

Analytic Jacobian matches FD (mid, band, calendar); fits agree	theorem 1	`models/lqd/jacobian.py`
		`test_lqd_jacobian.py::test_jacobian_matches_fd_mid`
		`::test_jacobian_matches_fd_band`
		`::test_jacobian_matches_fd_calendar`
		`::test_analytic_and_fd_fits_agree`

Barrier finite for wild trial parameters	(41)	`models/lqd/calibrate.py` `test_lqd_jacobian.py::test_barrier_residual_finite_for_wild_tail_scale`

Benchmark fit and speed rail (legacy fixture)	section 12	`models/lqd/calibrate.py`
		`test_lqd_calibrate.py::test_calibrate_to_svi_benchmark`
		`::test_calibration_is_fast_enough`

Double-hat event smile: bimodal density recovered	section 12.2	`models/lqd/calibrate.py` `test_lqd_pricing.py::test_double_hat_density_is_bimodal`
Calendar floor exact at constraint z-values	(50)	`calib/calendar.py` `test_lqd_jacobian.py::test_jacobian_matches_fd_calendar`

Endpoint/logistic charts: roundtrip, wall unreachable, chart-independent optimum (incl. a 12-node live SPY/NVDA/AAPL validation, $\lvert\Delta\theta\rvert_{\max} = 3.6 \times 10^{-7}$)	section 5.1	`models/lqd/charts.py`
		`test_lqd_endpoint_coords.py::test_logistic_chart_roundtrips_exactly`
		`::test_logistic_chart_cannot_reach_the_wall`
		`::test_logistic_fit_reaches_the_lr_optimum`

Numerical certificate: strike-space audit, Fritsch–Carlson margin, wing-rounding regression, overflow rejection, beyond-grid asymptote	section 7.4	`models/lqd/quadrature.py`, `interp.py` `test_lqd_numerical_certification.py` (whole battery)

A LQD-native controls and numerical constants

The knobs *specific to* the LQD slice, with defaults and roles. Surfaced knobs are user-facing `FitSettings`; hidden knobs are internal constants tuned to the algorithm and not exposed. The *shared* calibration controls that also shape an LQD fit — `fitMode`, `haircut`, `midAnchorWeight`, `enforceCalendar`/`calendarWeight`, the var-swap controls, and the prior-persistence and observation-filter settings of table 3 — are indexed in Note 00's control table and derived in their own notes (07, 08, 10, 13, 15); they are deliberately not duplicated here.

Table 6: LQD-native controls.

Knob	Default	Role
Surfaced (FitSettings)		
`nOrder` N	6	Legendre expansion order; 7 parameters at $N = 6$. Schema range $4 \leq N \leq 16$: 6 for ordinary slices, 8–14 for scheduled-event slices.
`regLambda` λ	10^{-6}	High-order ridge coefficient in (40); suppresses oscillation in sparse data.
`regPower` r	1.0	Ridge exponent in $n^{2r}a_n^2$; larger r penalizes high modes harder.
`barrierCenter` c	0.90	Centre of the A_R softplus barrier (41); engages before the $A_R = 1$ wall.
`barrierScale` γ	50.0	Steepness of that barrier.
`weightScheme`	`equal`	Per-quote weighting of the data residuals (`equal` or `tv_density`); see Note 07.
Hidden (internal constants)		
`Z_MAX` Z	40.0	Half-width of the symmetric log-odds grid; truncation error $\ll$ vega noise for equity tails.
`N_POINTS`	8001	Reporting/diagnostic grid nodes (odd, so $z = 0$ is a node).
`OPT_N_POINTS`	2001	Coarser grid used during optimization iterations; the accepted slice is rebuilt at `N_POINTS`.
`EPS_AR` ε_R	10^{-6}	Buffer in the hard bound $A_R < 1 - \varepsilon_R$.
`_VEGA_FLOOR` η	10^{-4}	Floor added to Black vega in the residual normalization (31).
`xtol/ftol/gtol`	10^{-10}	`trf` tolerances; ~ 6 orders below the fit budget, so the optimum is display-identical while `trf` stops grinding invisible reductions.
`max_nfev`	4000	Evaluation cap (rarely approached).

B Performance notes

1. **Cached static grid.** The grid $(z, u, u(1-u))$ and the Legendre matrix depend only on (Z, n_{pts}, N), so they are LRU-cached and returned read-only; only g is re-formed per call. This removed $\sim 40\%$ of `build_slice` and made each build $\sim 1.7\times$ faster.

2. **Two-grid optimization.** Iterations run on `OPT_N_POINTS` $= 2001$ (Simpson error already far below the fit budget); the accepted slice is rebuilt once at 8001 nodes for the reported result and diagnostics. Converged parameters agree with a full-grid fit to $\sim 10^{-6}$.

3. **Hermite interpolation/inversion.** Storing the exact nodal derivatives $\mathrm{d}Q/\mathrm{d}z$ and $\mathrm{d}G/\mathrm{d}z$

gives cubic-Hermite off-grid evaluation (clean Greeks) and machine-precision strike inversion by Hermite–Newton — and supplies precisely the nodal sensitivities the analytic Jacobian reuses.

4. **Analytic Jacobian** (section 10). One quadrature pass instead of $P + 1$ finite-difference rebuilds, reaching the same fitted solution within tolerance (relative cost agreement $\sim 10^{-11}$), measured 1.3–1.7× on the residual-path benchmark of table 7 and structurally more on heavier configurations. Gated to the var-swap/prior-free configuration.

5. **Looser tolerances.** Dropping `trf` tolerances from 10^{-15} to 10^{-10} stops the optimizer chasing cost reductions invisible in the priced surface, saving whole $(P{+}1)$-evaluation iterations at no display-precision cost.

The Jacobian timing measurement. Wall-clock speed-ups are machine-dependent, so the numbers promised in section 10 live here with their protocol stated in full — and one process rule adopted after an embarrassment: an earlier revision juxtaposed a timing table and a timing chart produced by *different generators on different days*, and their "fastest-of-three" numbers disagreed. Table and figure below are emitted by one generator from *one* measurement (`gen_lqd_audit.py`), and "fastest of k" is retired for the median with inter-quartile whiskers. Protocol: the running example's quote strip, mid fit, core residual blocks only; per order, one warm-up solve per Jacobian, then eleven *interleaved* analytic/FD solves (interleaving makes slow machine drift hit both estimators equally); single-threaded, warm static-grid cache; iteration counts reported because `trf` may take different step sequences under the two Jacobians. Different hardware, a cold cache, or an active var-swap/prior block will move the ratios.

Table 7: Analytic vs. two-point finite-difference Jacobian, one interleaved run of the production calibrator under the protocol above (median ± IQR over eleven solves each; "its." are iteration counts analytic/FD; the cost column confirms the same optimum within tolerance).

N	P	analytic (ms)	FD (ms)	speed-up	its. (a/FD)	$\lvert\Delta\text{cost}\rvert$
6	7	23.9 ± 2.1	39.2 ± 12.9	1.64×	8/15	2.6×10^{-16}
8	9	28.0 ± 7.2	40.4 ± 13.5	1.44×	8/9	7.1×10^{-17}
10	11	30.1 ± 2.5	47.5 ± 17.0	1.58×	8/9	8.7×10^{-18}
12	13	40.9 ± 21.8	80.5 ± 55.2	1.97×	7/8	1.3×10^{-17}

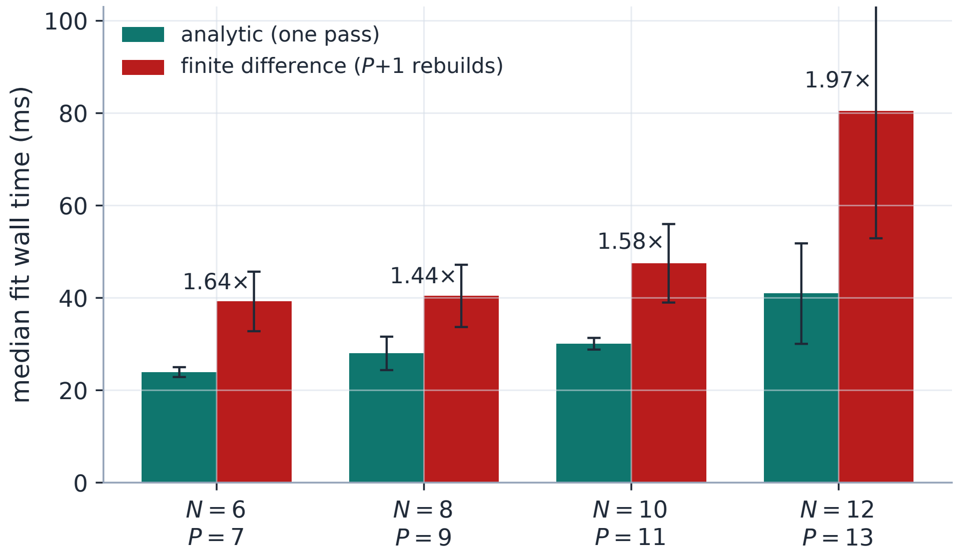

Figure 17: The same run drawn — by construction, since table and figure are emitted together: median per-fit wall time, IQR whiskers, ratio above each pair. The analytic pass is $O(P\,n_{\mathrm{grid}})$ (one cumulative sensitivity integration per parameter on one build), so its advantage over the $(P{+}1)$-rebuild finite difference is bounded, not unbounded, in P; the ratios' non-monotonicity in N tracks iteration counts, not noise.

Heuristic

Why only $\sim 2\times$ and not $\sim P\times$? Three reasons. The cached static grid makes a *repeat* `build_slice` cheap, so $P+1$ finite-difference rebuilds cost much less than $P+1$ cold builds. The analytic pass still pays $O(P\,n_{\mathrm{grid}})$ for its sensitivity integrals — it removes the rebuilds and root solves, not the linear-in-P work. And `trf` converges here in a handful of iterations, so linear algebra and the value evaluation dilute the Jacobian saving. The structural payoff — a Jacobian that no longer rebuilds the quadrature — is what makes higher-order and surface-wide fits comfortable.

C The synthetic oracles

The case files fit production against two deterministic quote oracles; neither is used anywhere inside the model.

The SSVI-shaped strip. The SPX-like target is the total-variance curve

$$w(k) = \frac{\vartheta}{2}\Big(1 + \rho\phi k + \sqrt{(\phi k + \rho)^2 + 1 - \rho^2}\Big), \qquad (\vartheta, \rho, \phi) = (0.0356,\, -0.68,\, 2.40), \tag{51}$$

Gatheral–Jacquier's SSVI slice shape [6], sampled at the 24 strikes listed in the generator and perturbed by the deterministic ripple $10^{-4}(1.20\sin(9k + 0.4) + 0.55\cos(21k))$ in volatility. (The regression fixture behind the legacy timing table, table 7, is the older five-parameter raw-SVI benchmark; the JW re-coordinatization used to specify it is Note 02's subject.)

The two-regime mixture. The event target is an equal-mean mixture of two log-return normals with weights $(56.0\%, 44\%)$, raw centres $(-0.075, +0.085)$ shifted by a common constant so the mixture is a martingale, and widths $(0.052, 0.047)$. Its normalized calls are the exact two-term Gaussian formula

$$C(k) = \sum_i w_i \Big(e^{m_i + \sigma_i^2/2}\, \Phi\Big(\tfrac{m_i+\sigma_i^2-k}{\sigma_i}\Big) - e^k\, \Phi\Big(\tfrac{m_i-k}{\sigma_i}\Big)\Big), \tag{52}$$

inverted to implied volatilities for quoting. This oracle has genuinely bimodal density, which is the point.

D Model card

One page, for a reader deciding whether to rely on this model without reading the lecture.

Object	One expiry's arbitrage-free marginal law of the log-forward return, parametrized by the log quantile density (5); $P = N + 1$ parameters, $N \in [4, 16]$.
Inputs	De-Americanized quotes (k_i, σ_i) in forward moneyness (forward hygiene is an input contract, Note 06); optional band edges, weights, calendar floor from a nearer expiry, var-swap and prior targets.
Outputs	Call/put prices, implied variance, density, CDF, exact ATM handles $(\sigma_0, s_0, \kappa_0)$, var-swap strike, tail scales and Lee slopes *with stability fans* (fig. 13).
Guarantees	Butterfly-freedom and parity structural per slice (proposition 2); implementation audited to residual arbitrage $\lesssim 10^{-8}$ of the forward at off-grid strikes (section 7.4); martingale error $\sim 10^{-16}$.
Controls	ATM retarget (exact); package controls (RR/BF/var swap, handle-neutral, with cross-talk and condition prints); shape modes (session-local); ridge λ, r; barrier; calendar toggle and weight; chart selection (logistic default; Gauss–Newton metric available).
Not guaranteed	Calendar order (controlled, not proven, section 11); tail point estimates (prior-dominated, fig. 13); any dynamics between expiries (section 14); atoms, gaps, super-exponential tails (proposition 1).
Failure modes	Interior overflow $\to$ clean rejection into the penalty branch; near-wall fits $\to$ barrier-dominated, conditioning degrades; retarget Newton can fail near the wall and reports failure; sub-vega-floor quotes fitted in price regime (fig. 7).
Approval scope	Marking liquid and moderately sparse equity smiles; research-grade tails. Equal-footing market benchmarks vs SVI-class fitters and hedge-P&L evidence live in the backtest programme and are *not* claimed here.

E Referee questions, answered where they live

This revision responds to a twelve-question committee review. Each answer is in the body; this index is the map.

1. *Why not endpoint-vanishing modes and a logistic A_R?* Both adopted and shipped: section 5.1.
2. *In what topology is the family dense, for what laws?* Proposition 1.
3. *How much can tails move with quotes inside spread?* Measured: fig. 13 (jackknife, order, ridge, noise and multi-start fans).

4. *What numerical arbitrage tolerance after interpolation?* Section 7.4: bounds 2.5×10^{-9}, butterflies 3.0×10^{-13}, digitals 3.7×10^{-12}.
5. *Worst-case quadrature error in* $(N, \theta, Z, 1-A_R)$*?* The audit's fine-grid comparison (2.9×10^{-9} across plain/near-wall/wild classes) plus the overflow budget; adaptive refinement remains future work.
6. *How many starts fail or split on real data?* Zero of ten, one basin, on both the synthetic and live strips (sections 9 and 12.1) — a measurement to repeat on stressed boards.
7. *What metric makes a handle move "cheapest"?* The chart now takes the Gauss–Newton metric of the fitted residuals (least fit impact), with the Euclidean default named as arbitrary and the Gram condition reported either way (section 8.2).
8. *Stable trader packages instead of kernel directions?* Shipped: kernel directions are session-local; RR/BF/var-swap package controls with cross-talk, rank and condition are the persistent interface (section 8.2).
9. *Deterministic fallback for overflow / wall / spikes / cap?* Overflow: budgeted rejection (section 7.2); wall: unreachable in the production chart, barrier-priced; spiky densities: ridge + judgment (section 14); evaluation cap: solver reports non-convergence.
10. *Live evidence vs SVI/eSSVI?* Not claimed here; backtest programme (section 14).
11. *Surface repair after an early expiry moves?* Screen-and-joint repair of violation-connected slices (section 11).
12. *Tail moments: approved outputs or diagnostics?* Diagnostics with mandatory fans (appendix D).

F Reference implementation

The whole slice is one quadrature pass in the log-odds coordinate: form g, integrate $\mathrm{d}Q/\mathrm{d}z = e^g$ to the quantile, solve one scalar normalization for the martingale shift μ, reverse-integrate the upper share G, and price every strike by (28). Executed against production before committing: the listing reproduces `build_slice`'s quantile and share to 10^{-12} and μ to 10^{-14} on an $N = 6$ slice, and returns $\mathbb{E}[e^X] = 1.000000$ (production inverts $Q(z_k) = k$ by Hermite–Newton and interpolates G with cubic Hermite for smooth Greeks; the linear `np.interp` here prices calls to $\sim 10^{-6}$ on the dense grid). Two wing-critical lines follow production's stable forms rather than the naive algebra — $u(1-u)$ as $\operatorname{expit}(z)\operatorname{expit}(-z)$ and the call's $e^k(1-u_k)$ leg in log space — because past $|z| \approx 36.7$ the naive forms round to zero against $u = 1$ (section 7.4); production additionally carries the interior overflow budget and the beyond-grid tail asymptote (32), omitted here as guard rails rather than pipeline.

Listing 2: The LQD quadrature pipeline (distilled from `models/lqd/{basis,quadrature}.py`).

```
def legendre(n_max, x):              # P_0..P_{n_max}, 3-term recursion
    P = np.empty((n_max+1, x.size)); P[0] = 1.0; P[1] = x
    for n in range(1, n_max):
        P[n+1] = ((2*n+1)*x*P[n] - n*P[n-1]) / (n+1)
    return P

def g_of_u(u, L, R, a):              # smooth part g(u) of log q(u)
```

```python
    return (1-u)*L + u*R + a @ legendre(len(a)+1, 1 - 2*u)[2:]

def build_slice(z, L, R, a):
    # One quadrature pass, eqs (Q_of_z)-(upper_share).
    u = expit(z); dz = z[1] - z[0]
    u1mu = expit(z) * expit(-z)    # u(1-u), exact into the wings
    # Tail scales e^{g(0)}, e^{g(1)}:
    A_L, A_R = np.exp(g_of_u(np.array([0.0, 1.0]), L, R, a))
    dQ = np.exp(g_of_u(u, L, R, a))           # dQ/dz = e^{g}
    Q = cumulative_simpson(dQ, dx=dz, initial=0.0)
    Q -= Q[len(z)//2]                         # anchor Q(0) = 0
    mass = np.exp(Q)*u1mu
    tot = (np.trapezoid(mass, z)              # + analytic tails
           + np.exp(Q[-1]-z[-1])/(1-A_R)
           + np.exp(Q[0]-z[-1])/(1+A_L))
    mu = -np.log(tot); Q = Q + mu             # now E[e^X] = 1
    m = np.exp(Q)*u1mu
    # Upper share G(z) = int_z^inf e^{Q} u(1-u):
    G = (cumulative_simpson(m[::-1], dx=dz, initial=0.0)[::-1]
         + np.exp(Q[-1]-z[-1])/(1-A_R))
    return u, Q, G

def call_price(k, z, u, Q, G):
    # C(k) = G(z_k) - e^k (1 - u_k), the last factor in log space
    zk = np.interp(k, Q, z)                   # invert Q(z_k) = k
    return np.interp(zk, z, G) - np.exp(k - np.logaddexp(0.0, zk))
```

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